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Simulating Grain Growth in PBF-LB/M with a Cellular Automaton Model to Determine Fracture Criticality

Simulation des Kornwachstums beim pulverbettbasierten Schmelzen von Metallen mittels Laserstrahl unter Verwendung eines Cellular-Automaton-Modells zur Risskritikalitätsbestimmung

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Abstract

In the last decade, the Additive Manufacturing (AM) process has become a popular method to produce components with a high degree of geometric complexity. Powder Bed Fusion of Metals using a Laser Beam (PBF-LB/M) has proven itself as a method to produce reliable metal components for uses requiring high strength and heat tolerance, but the prediction of mechanical properties, especially fracture behavior requires further work. The fracture mechanism depends on the microstructure, which is highly dependent on the process parameters used during the building process. To model this fracture behavior, a simulation is set up, based on 3D Cellular Automaton (CA) method widely used in literature, to model the mesoscale microstructure evolution during the build process. Methods to extract the relevant microstructural parameters were developed, and a fracture model incorporating this information devised. The microstructure simulation managed to model the grain growth for single layers but failed to demonstrate a key characteristic of the multi-layer microstructure: elongated grains extending across build layers in the build directions. A microstructure-aware fracture model was devised, based on the Johnson-Cook (JC) empirical fracture model shown to be suitable for PBF-LB/M, extending it to account for dependence on grain size and loading direction compared to large columnar grains. Due to limited experimental data and fracture experiments being out of the scope of this thesis, only a preliminary calibration of the grain size dependence was determined, and no calibration of the directional dependence was performed. Further work includes improved nucleation modeling, which is thought to impede the modeling of the grain growth across build, and experiments to quantify the parameters of the extended fracture model. Developing a Crystal Plasticity Finite Elements Method (CPFEM) fracture simulation incorporating the results of the microstructure simulation should also be considered.

Zusammenfassung

In den letzten Jahren ist das Verfahren der additiven Fertigung (AM) zu einer beliebten Methode zur Herstellung von geometrisch komplexen Bauteilen geworden. Das pulverbettbasierte Schmelzen von Metallen mittels Laserstrahl (PBF-LB/M) hat sich als Methode zur Herstellung zuverlässiger Metallbauteile für Anwendungen bewährt, die eine hohe Festigkeit und Hitzetoleranz erfordern. Dennoch, die Vorhersage der mechanischen Eigenschaften, insbesondere des Bruchverhaltens, muss besser verstanden werden. Der Bruchmechanismus hängt von der Mikrostruktur ab, die mit den im Bauprozesses verwendeten Prozessparametern stark variieren kann. Um dieses Bruchverhalten zu modellieren, wird eine Simulation mithilfe der in der Literatur verbreiteten 3D-CA-Methode erstellt, um die Entwicklung der Mikrostruk-

tur auf der Mesoskala während des Bauprozesses zu modellieren. Methoden zur Extraktion der relevanten mikrostrukturellen Parameter wurden entwickelt, und ein Bruchmodell erstellt das diese Informationen ausnutzt. Mit der Mikrostruktursimulation gelang es, das Kornwachstum für Einzelschichten zu modellieren, aber ein wesentliches Merkmal der mehrschichtigen Mikrostruktur fehlte: kolumnare Körner, die sich über die Schichten in den Aufbaurichtungen erstrecken. Es wurde ein mikrostruktur-abhängiges Bruchmodell entwickelt, das auf dem empirischen Johnson-Cook (JC) Bruchmodell basiert, welches für PBF-LB/M schon als geeignet erwiesen wurde. Dies wurde erweitert um eine Abhängigkeit der Bruchfestigkeit von der Korngröße und der Belastungsrichtung relativ zur Ausrichtung der kolumnaren Körner. Aufgrund begrenzten experimentellen Datens, und dass Bruchexperimente nicht im Rahmen dieser Arbeit vorgesehen sind, wurde nur eine vorläufige Kalibrierung der Korngrößenabhängigkeit bestimmt und keine Kalibrierung der Richtungsabhängigkeit durchgeführt. Weitere Arbeiten umfassen eine verbesserte Modellierung der Keimbildung, da man annimmt, dass durch der jetzigen Implementierung das kolumnare Kornwachstum behindert wird, sowie Experimente zur Quantifizierung der Parameter des erweiterten Bruchmodells. Die Entwicklung einer CPFEM-Bruchsimulation unter Einbeziehung der Ergebnisse der Mikrostruktursimulation sollte ebenfalls in Betracht gezogen werden.

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List of Abbreviations

AM	Additive Manufacturing
Ascent	Ascent AM Simulations
BD	Build Direction
CA	Cellular Automaton
CCX	CalculiX CrunchiX
CET	Columnar to Equiaxed Transition
CFD	Computational Fluid Dynamics
CFL	Courant–Friedrichs–Lewy
CPFEM	Crystal Plasticity Finite Elements Method
CTOD	Crack Tip Opening Displacement
EPFM	Elastic-Plastic Fracture Mechanics
FCG	Fatigue Crack Growth
FEM	Finite Elements Method
IN718	Inconel 718
IPF	Inverse Pole Figure
JC	Johnson-Cook
LBM	Lattice-Boltzmann Method
LEFM	Linear-Elastic Fracture Mechanics
ParaView	The ParaView Visualization Package
PBF	Powder Bed Fusion
PBF-LB/M	Powder Bed Fusion of Metals using a Laser Beam
PDE	Partial Differential Equation
PF	Phase Field
RT	Rice-Tracey
SEM	Scanning Electron Microscope
SPH	Smoothed-Particle Hydrodynamics
VED	Volumetric Energy Density

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Chapter 1

Introduction

1.1 Motivation

The AM process has found great success in the last decade in various industries as a cutting-edge production technique. PBF-LB/M is one of the most promising methods for applying this paradigm to metals and has been shown to produce strong and reliable parts with highly complex geometries. (SANCHEZ et al. 2021) The capability of producing lightweight structural parts from metals with high strength and melting points makes it an attractive method for producing critical components such as propellers and engine pistons. (SAMUEL et al. 2018) For these applications, it is essential to have highly informed estimates of the fracture behavior, as cracking could have devastating effects. Experiments have shown that fracture behavior highly depends on the microstructure, knowledge of which currently is not incorporated into fracture estimates.

The microstructure of AM parts, contrary to conventional production techniques such as forging or casting, is highly variable. It depends strongly on the process parameters during the building process, with few available heuristics for predicting the resulting microstructure. (KURZ et al. 2023) Were it necessary to investigate the microstructure or fracture behavior of each part for every new set of process parameters used, the flexibility and usefulness of AM would decrease drastically. Additionally, varying the process parameters during the build process has been shown to provide desired mechanical properties, but would necessitate even more testing to guarantee structural reliability. (BRENNE and NIENDORF 2019) Predicting the fracture behavior dependent on the microstructures resulting from various process parameters and geometries would greatly aid the development of new process adjustments to enhance the mechanical properties, and simplify the necessary verification of process parameters when producing critical parts. (SAMUEL et al. 2018)

1.2 Objective

To enhance the estimates of the fracture behavior for PBF-LB/M manufactured parts, a simulation model for the microstructure evolution during the building process is implemented, and a fracture model is devised to model the dependence of the fracture toughness on the simulated microstructure. The necessary steps to achieve this are outlined in the following section, providing a structure of the thesis.

1.3 Structure

Chapter 2 treats the background knowledge informing the methods employed in the thesis. First, the PBF-LB/M manufacturing technique is described before giving a short introduction to fracture mechanics to provide context to insights on the microstructure-influenced fracture behavior. An introduction to the solidification theory and the mechanisms observed to influence the microstructure formed during PBF-LB/M are described, before presenting currently available methods for simulating the microstructure. As the chosen microstructure simulation method relies on knowledge of the temperature fields, methods for modeling and simulating these are also presented.

The fundamental knowledge and state-of-the-art methods presented in Chapter 2 are combined in Chapter 3 to present the methods employed to simulate the microstructure, along with methods for verifying and validating the microstructure simulation. Methods to extract information quantifying microstructural parameters are also detailed and used to inform the microstructure-aware fracture model ultimately presented.

Subsequently, in Chapter 4, the results of the verification and validation of the microstructure are presented, as well as the results obtained with the fracture model. The limitations of the used models and the obtained results are also discussed.

Finally, the method and findings are summarized in Chapter 5, along with an outlook on further work that could build upon the results from this thesis and alternative methods that should be investigated to alleviate the limitations posed by the here employed methods.

Chapter 2

Fundamentals

2.1 Power Bed Fusion with Laser Beam Melting

AM, commonly described as 3D printing, is a novel method of producing objects layer by layer from a digital model. This technique stands out for its ability to create complex geometries often infeasible with traditional manufacturing methods, or with a drastic reduction in material waste and costs. In the last decade, it has increasingly been used for metallic materials, and one of the most promising processes is PBF-LB/M, illustrated in Fig. 2.1. The process uses a high-power laser beam, fusing fine metallic powders into solid parts layer by layer, directly determined by a digital input geometry. Whilst providing great opportunities for material utilization, precision, and interwoven geometries impossible to realize by casting, the process is based on the interaction of several competing physical phenomena that have to be understood well to produce durable and reliable parts. (MEIER et al. 2017) This section will outline the key process steps in PBF-LB/M, separated into pre-processing, the build process, and post-processing.

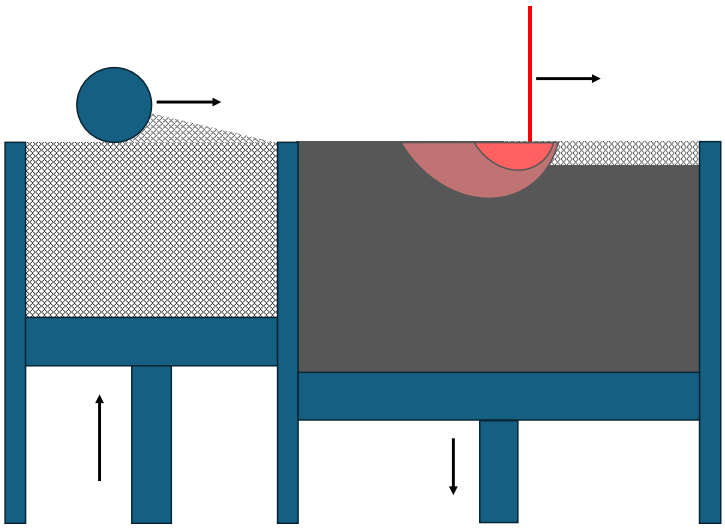


Figure 2.1: A schematic overview of the functioning of PBF-LB/M. Adapted from MEIER et al. (2017)

2.1.1 Pre-Processing

Pre-processing contains everything before the build process, including modeling a 3D geometry, optimizing the geometry or process parameters, slicing the geometry into layers with the appropriate laser hatching tracks, and setting up the machine with the desired process settings.

All AM methods require a 3D geometry as an input, before slicing this into appropriate layers for manufacturing. Whilst the design phase can enjoy the added creative freedom from the realizable geometries, they typically cannot take advantage of the full potential of AMs capabilities regarding material savings. Optimizing the geometry of a part commonly involves topology optimization, where 3D models can be optimized to use less material, and thus be lighter and cheaper than conventionally produced parts. This is done by only defining the part's physical boundary conditions and structural demands, and allowing an optimization algorithm to develop complex geometries that fulfill optimized to some constraint, typically to minimize material use. (GIBSON et al. 2015) The geometry may also require structural supports to be built along with the desired main part, the need for which can be determined using structural calculations on a finalized geometry.

For the build process itself, many parameters can be adjusted in PBF-LB/M, of which several are known to significantly influence the resulting mechanical properties. These can generally be split into laser-related, scan-related, powder-related, or temperature-related parameters. (CHOWDHURY et al. 2022)

To simplify the choice of parameters the thermodynamic quantity Volumetric Energy Density (VED) is commonly employed:

$$VED = \frac{P}{v\tau\sigma} \text{ [J/mm]} \tag{2.1}$$

v being the scan velocity of the laser in mm/s, τ the layer thickness in mm and σ the laser spot diameter in mm. (CIURANA et al. 2013)

Physical simulations are an essential part of this step, as the prediction of mechanical properties based on process parameters has few analytical or heuristic approaches to rely on (KURZ et al. 2023). Several research projects are developing complete simulation packages for the whole build process, such as BIERWISCH et al. (2022), which developed a process chain to simulate the powder layer deposition upon the bed, the radiative heat transfer, and fluid dynamics of the melt pool, the microstructure evolution, and the resulting mechanical properties. All these simulation steps employ different physical and computational models, and methods have been developed for differing computational costs, simulation sizes, and levels of physical detail.(ZINOVIEVA et al. 2023) Methods commonly employed for these simulations are outlined in sections 2.4 and 2.5.

2.1.2 Build Process

During the build process, thin layers of metal powder are spread across the build area using a roller before a high-powered laser impinges upon a small section of the powder. The laser melts deeper than the applied powder layer, fusing it with existing material below. It also produces a melt pool wider than the laser beam as the heat spreads. Scanning the powder bed using a predetermined hatching strategy, the laser fuses the powder layers together using scans sufficiently close that portions of the previously melted powder will be remelted, resulting in strongly bonded structures within and between layers. (GIBSON et al. 2015)

As seen in Fig. 2.1, the build platform is typically lowered by the layer height, whilst a powder reservoir supplies the roller with sufficient new powder, this process is repeated for every layer. Commonly the bottom of the part is rigidly attached to the bottom of the build platform to reduce distortions. The process typically also includes a significant warm-up and cool-down time for the machine, where producing large parts is not significantly more time-consuming. (GIBSON et al. 2015)

2.1.3 Post-Processing

Encompassing all process steps after the laser has finished melting all layers, post-processing typically encompasses stress annealing the part to remove residual stresses, separating the part from the build area, machining away surface defects and structural supports, and heat treatment to improve mechanical properties. As parts enjoy natural structural support by surrounding powder during the building process, this remaining powder has to be removed after processing. Often, additional structural support is necessary; this has to be machined away. (GIBSON et al. 2015) Ideally, very little post-processing is necessary, however, this depends on using suitable process parameters that avoid deformations and satisfy the requirements for precision and surface roughness. (STUCKER 2011) Otherwise, mechanical surface treatment or heat treatment may be required. (YU et al. 2020) (ZINOVIEVA et al. 2023)

2.2 Fracture Mechanics

DOWLING (2013) defines the fracture of a material as: "Cracking to the extent that a component is separated into two or more pieces". This gives an important clue to the two separate aspects of a fracture. Firstly, an initial fault in the material must appear to start a fracture, followed by conditions favouring the propagation of this crack all the way through the material until it is split into two or more pieces. (ANDERSON 2017) These two aspects are typically called crack initiation (or nucleation) and crack propagation. In this thesis, the scope will be limited to fractures in ductile metals, as primarily the nickel alloy Inconel 718 has been investigated, which falls within the category. First, fracture mechanics theory and the process of ductile fracture will be introduced based on ANDERSON (2017) before considering microstructural influences found specifically in AM metals.

2.2.1 Theoretical Background

On a microscopic level, a fracture necessitates that the inter-atomic bond is broken between a material's neighbouring atoms. Applying external energy and stress to a fracture can overcome the bond energy and propagate it further. An essential concept here is that the local stress at a small crack in a material will concentrate the stresses, and cause infinitely high local stress for an infinitesimal crack. This constitutes the basis of the critical stress criterion in Linear-Elastic Fracture Mechanics (LEFM). Using this criterion, one can estimate local stress as a function of the globally applied stress, with a crack width assumed to be in the order of magnitude of the atomic radius of the material.

This was extended to satisfy energy conservation:

$$\frac{\mathrm{d}E}{\mathrm{d}A} = 0, \tag{2.2}$$

with Griffith's energy criterion postulating that the change in potential strain energy Π and the work needed to create new surfaces W_s needs to be equal.

$$\frac{\mathrm{d}E}{\mathrm{d}A} = \frac{\mathrm{d}\Pi}{\mathrm{d}A} + \frac{\mathrm{d}W_s}{\mathrm{d}A} \implies \frac{\mathrm{d}\Pi}{\mathrm{d}A} = -\frac{\mathrm{d}W_s}{\mathrm{d}A}. \tag{2.3}$$

To simplify the use in an engineering context, this is commonly reformulated to the energy release rate

$$\mathcal{G} = -\frac{\mathrm{d}\Pi}{\mathrm{d}A} = R, \tag{2.4}$$

determining how much energy is available for an extension of the crack. Another measure for the crack resistance R is introduced, giving a condition for the state where a fracture will propagate. Commonly the stress intensity factor K is determined to quantify the increase of the stress due to a given crack length a, defined as

$$K = Y\sigma\sqrt{\pi a}\,, \tag{2.5}$$

where Y is a dimensionless geometry factor for the crack and its surroundings, and σ is the applied stress.

LEFM is only valid for linear elastic materials obeying Hooke's law, meaning they exhibit a proportional and linear relationship between stress and strain DOWLING (2013). For real

engineering materials, especially ductile metals, the estimates can be off by several orders of magnitude, and other approaches have been developed within Elastic-Plastic Fracture Mechanics (EPFM). One explanation for this commonly used method is the J-integral

$$J = -\frac{\mathrm{d}\Pi}{\mathrm{d}A} = -\frac{\mathrm{d}(U-F)}{\mathrm{d}A}, \tag{2.6}$$

as a more general extension of the energy release rate $\mathcal{G}$ also valid for elastic-plastic materials. This approach models elastic-plastic materials as nonlinear elastic and redefines Π as $U-F$, U being the strain energy stored in the body, and F as the external forces' work. Setting $F = 0$, in Eq. (2.6), as the work in an ideally elastic material is always zero if there is no displacement, Eq. (2.4) from LEFM is retrieved.

To describe the whole fracture behaviour of a material, or a material probe, fracture resistance curves such as the $J-R$-curve are commonly used to give an impression of the overall fracture behaviour. Plotting the energy release rate J against the length of the crack extension. In ductile metals a plastic deformation occurs around the crack's tip occurs right after crack initiation. This increases the radius of the crack to a size δ, and thereby also the fracture resistance by several orders of magnitude. The progress of this crack tip blunting is seen qualitatively above the curve in Fig. 2.2. The size δ_t:

$$\delta_t = \frac{4\mathcal{G}}{\pi\sigma_{YS}}, \tag{2.7}$$

can be obtained as a function of the energy release rate $\mathcal{G}$ and the yield stress σ_{YS} by Eq. 2.7 according to the commonly used Crack Tip Opening Displacement (CTOD) theory, which can be viewed as a special case of the J-integral.

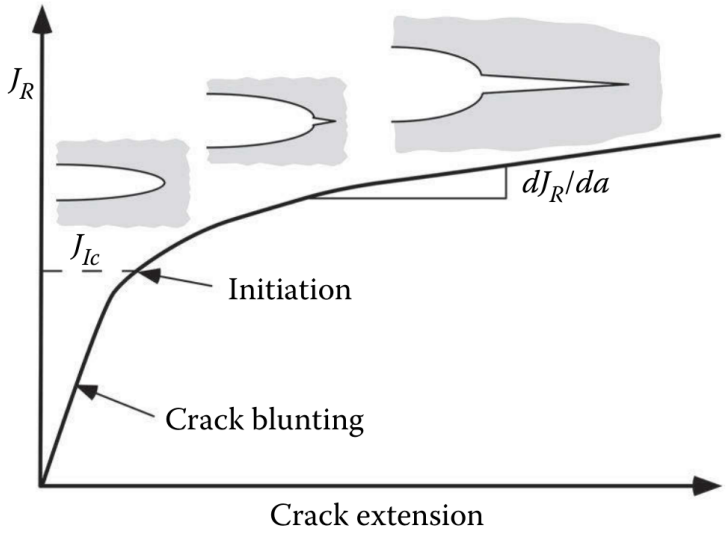
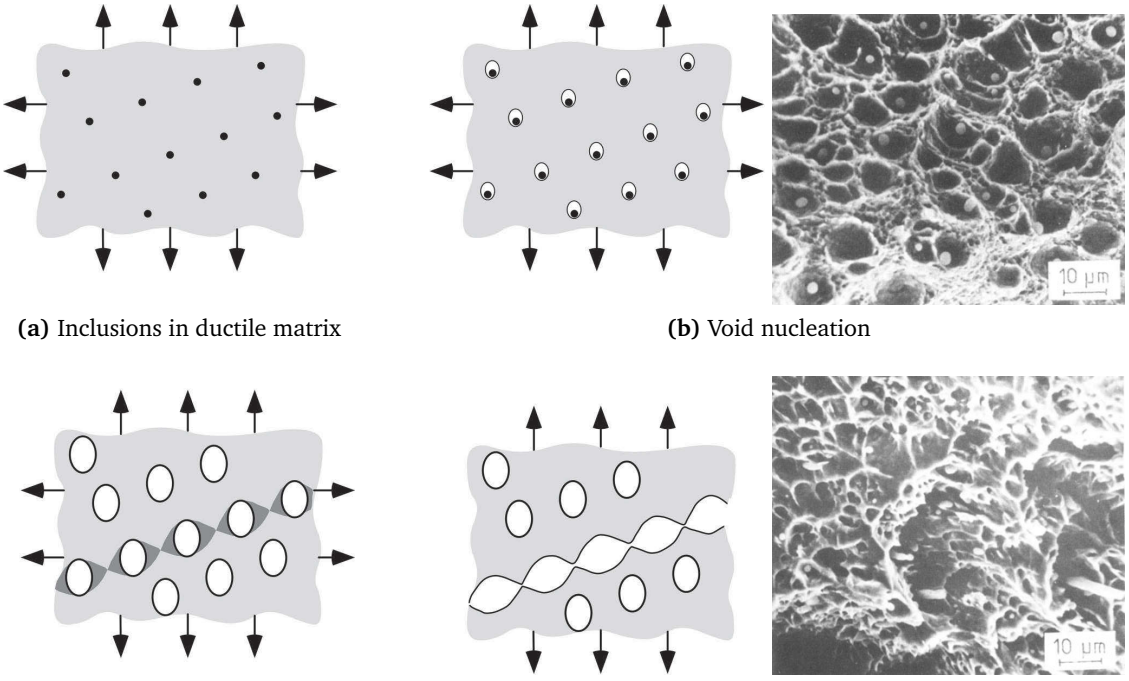


Figure 2.2: Schematic J resistance curve for ductile material ANDERSSON and LUNDBÄCK (2023)

2.2.2 Stages of Ductile Fracture

The following observations are only valid for the case of ductile fracture, the dominant fracture mode in the first stages of cracking in ductile metals. Ductile fracture can be separated into three distinct stages, illustrated in Fig. 2.3:

1. Void nucleation: Formation of a free surface around a particle by interface decohesion or particle cracking
2. Void growth: Growth of a void around the particle due to plastic strain and hydrostatic stress
3. Ductile crack growth: Coalescence of growing voids into a crack



(a) Inclusions in ductile matrix

(b) Void nucleation

(c) Void growth and strain localisation between voids

(d) Void coalescence and fracture

Figure 2.3: The stages of ductile fracture, including an initial stage before initiated fracture in (a). Images from SEM. Material 28 NiCrMoV 85 in (b), and CrNi l8/8+S in (d). Adapted from ANDERSSON and LUNDBÄCK (2023) and TSCHEGG (1982)

Void Nucleation

Void nucleation typically occurs around secondary particles not part of the material matrix. While not all these secondary particles induce void nucleation, the cohesion between matrix and particle is considerably weaker than within the matrix, increasing the chance of void nucleation. Given the mean or hydrostatic stress σ_m, and the effective von Mises stress σ_e, a measure for the critical stress σ_c at which interface decohesion occurs can be determined.

$$\sigma_m = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3) \tag{2.8}$$

$$\sigma_e = \sqrt{\frac{(\sigma_1 - \sigma_2) + (\sigma_1 - \sigma_2) + (\sigma_1 - \sigma_2)}{2}} \tag{2.9}$$

$$\sigma_c = \sigma_m + \sigma_e \tag{2.10}$$

While this macroscopic view can sometimes give the desired results, there are several approaches based on it that often come closer to the desired results. What is certain is that ductile fractures occur more readily when there is a high stress-triaxiality σ^*, when the hydrostatic stresses dominate the deviatoric stresses, which promotes the formation of voids.

$$\sigma^* = \frac{\sigma_m}{\sigma_e} \tag{2.11}$$

Void Growth

Two different fracture models will be introduced, shown by PANZER, WOLF, et al. (2023) to be suitable for ductile metals produced using AM; the Rice-Tracey (RT) (RICE and TRACEY 1969) and JC (JOHNSON and COOK 1985) fracture models. These are both strain-based failure models, where the critical strain is a function of the stress state. They assume a single initial spherical void in the material that is deformed non-uniformly, and attempt to establish a semi-empirical relationship for the growth velocities $\dot{r_1}, \dot{r_2}$ and $\dot{r_3}$ of the deforming ellipsoid.

D'ESCATHA and DEVAUX (1979) built upon the original paper by Rice and Tracey and found an empirical result valid for ductile metals that experience strain hardening,

$$\ln\left(\frac{\bar{R}}{R_o}\right) = 0.283 \int_0^{\varepsilon_{eq}} \exp\left(\frac{3\sigma_m}{2\sigma_e}\right) d\varepsilon_e^p \,, \tag{2.12}$$

where $\bar{R}$ is the mean radial displacement of the ellipsoid, and $d\varepsilon_e^p$ is the effective plastic strain increment, defined as:

$$d\varepsilon_e^p = \sqrt{\frac{2}{9}\left(\left(d\varepsilon_1^p - d\varepsilon_2^p\right)^2 + \left(d\varepsilon_2^p - d\varepsilon_3^p\right)^2 + \left(d\varepsilon_3^p - d\varepsilon_1^p\right)^2\right)} \,. \tag{2.13}$$

This can be reformulated using a material and process-dependent calibration parameter α_{RT},

$$\int_0^{\varepsilon_e^f} \exp\left(\frac{3\sigma_m}{2\sigma_e}\right) d\varepsilon_e = \alpha_{RT} \,. \tag{2.14}$$

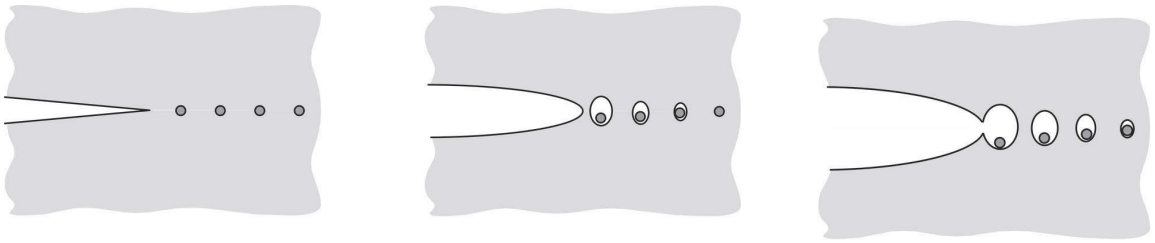
This can be interpreted as a fracture damage accumulation model, where α_{RT} is the total damage tolerated before crack initiation, ε_e^f being the strain at which this occurs.

The JC model extends this semi-empirical formula, to a purely empirical model with several additional calibration parameters. The concept is to extend the original equation by multiplicative factors to adjust for different physical effects, defining ε_e^f, the strain at which failure occurs, as:

$$\varepsilon_e^f = (d_1 + d_2 \exp(d_3\sigma^*))\left(1 + d_4 \ln \dot{\varepsilon}^*\right)(1 + d_5 T^*), \tag{2.15}$$

$$\dot{\varepsilon}^* = \frac{\dot{\varepsilon_e^p}}{\dot{\varepsilon_{ref}}}, \qquad T^* = \frac{T - T_{ref}}{T_{melting} - T_{ref}} \,, \tag{2.16}$$

T^* being the dimensionless temperature during loading, and ε^* the dimensionless effective plastic strain rate, and σ^* the stress-triaxiality from Eq. 2.11. d_1, d_2 and d_3 are additional



(a) initial state **(b)** void growth at crack tip **(c)** coalescence of voids at crack tip

Figure 2.4: Stages of ductile crack propagation. Adapted from ANDERSSON and LUNDBÄCK (2023)

degrees of freedom to account for effects such as initial plastic flow before voids have nucleated.

This approach allows for simple extension with other physical factors that may be of interest, for example, information about the very heterogeneous microstructure typically found in metals produced using PBF-LB/M.

Crack propagation

The last important aspect of ductile fracture is the propagation of cracks. Crack propagation can typically be viewed as repeated crack initiation, with different geometries reflecting the extending crack, applying the same criteria as above. On a microstructural level, this means inducing the nucleation, growth and coalescence of new voids in front of the crack tip, if there is sufficient stress. While this could theoretically be applied analytically, this incremental view on crack propagation is particularly well suited for computational methods. ANDERSON 2017

Most numerical methods for fracture are based on FEM, as it is very well suited for calculations of internal stresses and strain. However, it has to be extended to handle changing geometry and discontinuities as cracking occurs. PANZER (2020) realized this using the Element Death Method, where the elements where the fracture has occurred are deactivated further FEM calculation. As this requires a high density of elements, meaning many mesh refinements where the fracture occurs, a variety of more sophisticated XFEM approaches also exist where mathematical discontinuities are handled directly, but for the sake of brevity, these will not be discussed further here. (YAZID et al. 2009)

2.2.3 Influences of Microstructure on Fracture

A key observation motivating this thesis is the significant influence the microstructure of a metal can have on its fracture properties. As the microstructure of AM materials depends strongly on its process parameters, the fracture behavior is also very heterogeneous. (PANZER et al. 2024) The behavior of cast and forged material, with more a predictable microstructure, has been extensively investigated in the 70s to 90s, such as by MILLS and BLACKBURN (1990). Still, these results are not necessarily applicable to AM metals (VIEILLE et al. 2020).

The bulk of data available for fracture behaviour in AM materials correlates process conditions such as hatching strategy (VIEILLE et al. 2020) and (BRENNE and NIENDORF 2019), laser power, or scan speed (PANZER et al. 2024) directly with fracture. However, by investigating the resulting microstructure experimentally, a link between the microstructure and fracture

can also be inferred.

BRENNE and NIENDORF (2019) studied 316L Stainless Steel and showed that fractures can be halted at the boundaries between crystal grains in the metal if the intercrystalline crack growth in a direction non-parallel to the existing crack exhibits lower resistance than a further extension of the existing crack, as seen in Fig. 2.5. They attributed this to lowered effective stress σ_e at the tip of a deflected crack, preventing the initiation of further cracks. This deflection was achieved for a crack tilted 45° to the BD, where elongated grains aligned with the building direction were formed, the mechanism of which is further detailed in section 2.3.3.

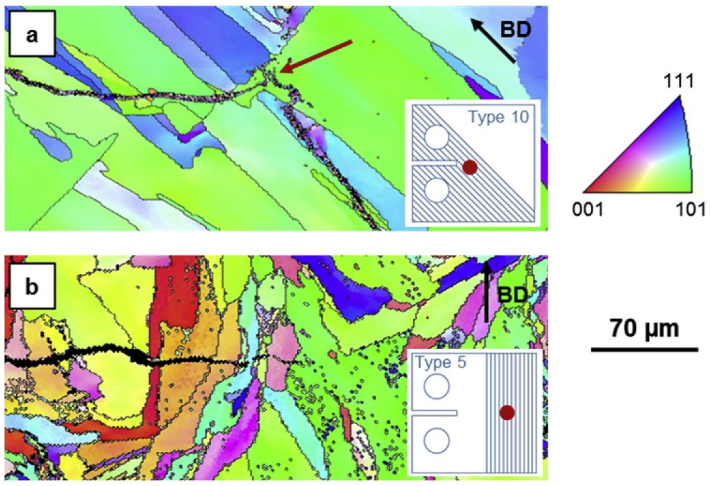
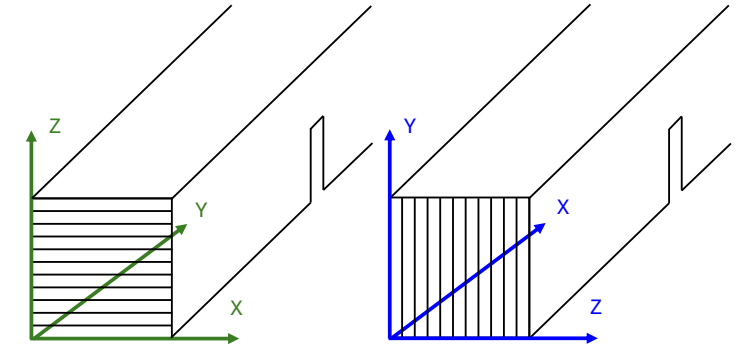
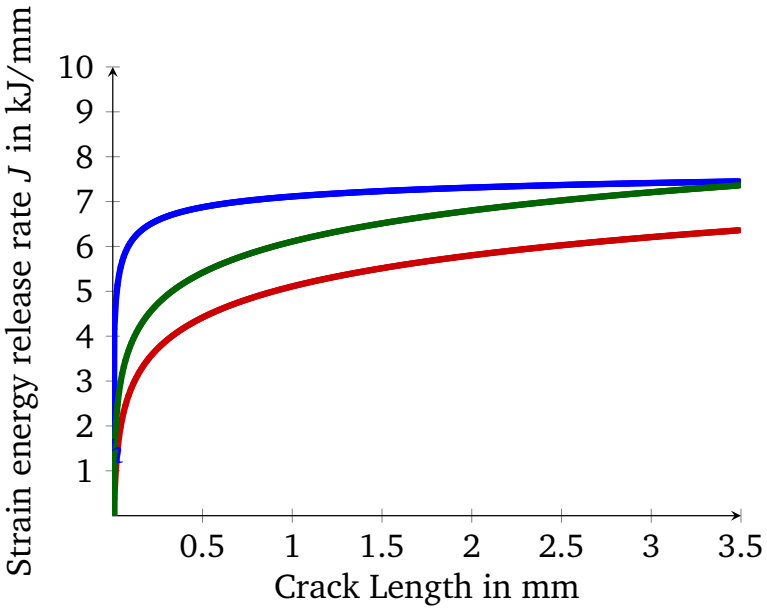


Figure 2.5: IPF showing the crack development and grain directions for two specimens printed with differing hatching patterns. The crack in a) was halted as it was deflected away from the direction of crack propagation, and the effective stresses were reduced; the crack in b) did not experience this and cracked further into the specimens. (as seen in the bottom-right sub-diagram) From BRENNE and NIENDORF (2019)

VIEILLE et al. (2020) performed single-edge notch bending tests on IN718 and showed that AM parts could demonstrate a dramatically higher fracture criticality than comparable specimens formed by casting. However, they could feature a very different J-R fracture curve more common in brittle materials, with a rather flat R curve rather than linear ductile tearing, as seen in Fig. 2.2 This was more pronounced for specimens with BD perpendicular to the crack growth direction, where they attributed this to the fact that after the onset of fracture, the new fracture surfaces were considerably larger in area than for the cast specimens and the AM specimens with the BD perpendicular to the crack growth direction. For further data in this field, it is necessary to understand the solidification and evolution of microstructure to obtain desired microstructures for testing and application, which will be discussed in Section 2.3.



(a) Laser hatching direction **(b)** Laser hatching direction



(c)

Figure 2.6: Schematic J-R fracture resistance curve based on VIEILLE et al. (2020), where red corresponds to cast specimens, and green and blue for corresponding laser hatching and build directions illustrated in (a) and (b), where Z is the build direction.

2.3 Solidification of Metals

One of the main motivations for melting metal is the dramatic reduction of viscosity between solid and liquid states by around 20 orders of magnitude and the highly simplified shaping by this process, however, this solidification significantly impacts the material properties. In this chapter, the core parts of metal solidification will be explained, mainly based on KURZ et al. (2023), before expanding with special considerations needed for the solidification phenomena found in additive manufacturing.

Unlike elements like polymers or glasses, the solidification of metals centers around the formation and growth of solid crystals within the melt until all metal is solidified. The speed at which the solidification occurs, the heat flow directions within the solidifying metal, and its elemental composition greatly affect the microstructure formed by this process. Any defects formed in the metal during solidification are difficult to remove and may remain a weak spot even after extensive reshaping of the resulting solid metal.

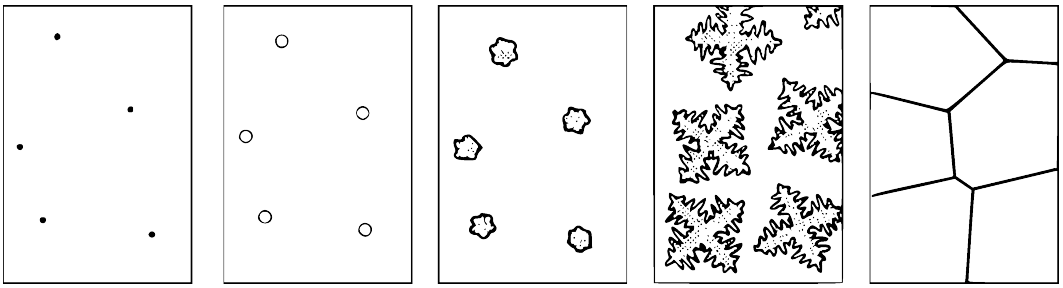


Figure 2.7: Progress of equiaxed solidification in dendritic growth. Adapted from KURZ et al. (2023)

For solidification to occur, the melt's temperature has to be undercooled at some place, i.e., having a temperature below the T_{liquidus} of the metal. This allows the nucleation of solid crystals within the melt. Dendrites grow out from these new crystals in all directions with velocities dictated by the local temperature gradient and ultimately grow together to form large grains with grain boundaries where the different crystals meet. As seen in Fig 2.7, once these grains have formed, the structure of the dendrites becomes invisible, and can only seen by the varying local composition of the alloys due to microsegregation. This is different for eutectic and peritectic metals, but pure metals and complex alloys (such as Inconel 718) typically have no stable compounds that form during solidification to change this behaviour.

2.3.1 Nucleation

An essential first step in determining the microstructure is modelling the nucleation process accurately, as the density of grains in the final microstructure is determined early in the solidification process by how many nucleations occur, as illustrated in 2.7. Only once the volume is dominated by solids does the crystal growth control the further development of the microstructure. As nucleation necessitates a formation of new surfaces, with a corresponding curvature at the solid-liquid interface, as this curvature is related to the melting point of the nucleus, a minimum grain size at nucleation r° can be determined by as:

$$r^\circ = \frac{2\Gamma}{\Delta T_\mathrm{r}}\,, \tag{2.17}$$

where ΔT_r is the change in melting point due to this curvature, and Γ is the Gibbs-Thompson coefficient.

The critical condition for nucleation to occur, can be expressed in terms of solid-liquid interface energy σ and Δg, the Gibbs free energy difference between solid and liquid.

$$\Delta G^\circ_{n,hom} = \frac{16\pi}{3}\frac{\sigma^3}{\Delta g^2} \tag{2.18}$$

This equation is only valid for homogeneous nucleation with only liquid metal melt, featuring no interactions with existing solid surfaces. The modeling can be extended to where the critical condition can be expressed as a simple extension of 2.18 accordingly:

$$\Delta G^\circ_{n,het} = s(\theta_c)\Delta G^\circ_{n,hom}\,, \qquad s(\theta_c) = \frac{(2+cos\theta_c)(1-cos\theta_c)^2}{4}\,, \tag{2.19}$$

where θ_c is the solid-solid wetting angle between the new nucleus and an existing solid. While this difference might seem negligible, it significantly impacts the nucleation rate I, appearing as an exponential term:

$$I \propto exp\left(-\frac{\Delta G_n^\circ + \Delta G_d}{k_B T}\right), \tag{2.20}$$

k_B being the Boltzmann constant, and ΔG_d the activation energy of the atoms passing from liquid to solid.

2.3.2 Growth

Once nucleated, there are two important different modes of dendritic crystal growth; columnar, or epitaxial, and equiaxed growth. Which growth mode applies to a grain is mainly determined by the local heat flow, from which it can obtain enough energy for the fusion of new dendrite surface area. Unconstrained, the growth mode is equiaxed, as illustrated in Figure 2.7 where the grains grow in all directions equally. However, when there is competition amongst grains and a clear direction of the heat flow, the growth is faster in one direction and limited in the others, causing elongated grains to grow anti-parallel to the direction of the heat flow. This growth is clearly dendritic, reflecting the preferential growth direction of the crystal lattice. In cubic crystal lattices, this is the crystallographic direction $\langle 100 \rangle$. It is important to consider that a sufficiently high grain density inhibits the formation of dendrites, leading to purely equiaxed growth zones. KURZ et al. (2023) (p.19)

To exemplify these two growth modes, consider a typical metal casting with a mould that is significantly cooler than the molten metal that has just been poured into the mould. As seen in Fig 2.8, an outer equiaxed zone forms along the edge of the mould in the beginning as this is where the melt undercools first, followed by columnar growth as the heat flow clearly goes from the centre of the melt to the mould, and there is a lack of space for the grains to grow freely. Further away from the mould, the heat flow is less directional, and the radial heat flow from the undercooled melt causes an inner zone of equiaxial growth, around detached dentrite tips from the columnar zone.

The dendrite tip velocity is a key parameter for what growth mode is occurs, and is a fairly well understood concept with a good theoretical framework in the Ivantsov function relating diffusion, undercooling and the dendrite tip velocity. It is however an implicit equation too costly for implementation in simulations, a problem which lends itself to approximation. A model for low Peclet numbers ($Pe < 0.1$), where diffusion dominates the heat transport, was found by RAPPAZ and DANTZIG (2009). $k, \Delta T_0$ and D being material constants, and $\Delta T = T_{liquidus} - T$ the local undercooling.

$$V_d = \frac{1}{5.51\pi^2} \frac{D}{\Gamma(k\Delta T_0)^{1.5}} \tag{2.21}$$

While few statements can be made in general about the influence microstructures have on mechanical properties, a general trend is that finer microstructures have superior mechanical properties. This is due to grain boundary strengthening, where grain boundaries prevent or deflect dislocation due to different orientations of the crystallographic planes between neighbouring grains. (DOWLING 2013)

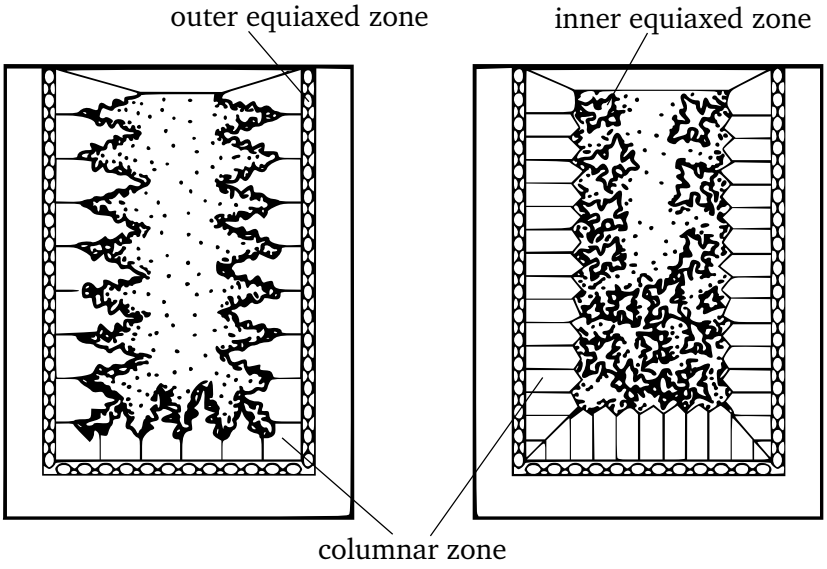


Figure 2.8: Growth zones in casting. Adapted from KURZ et al. (2023)

2.3.3 Rapid Solidification in Additive Manufacturing

Finer microstructures stem from higher cooling rates, leading to higher grain nucleation and growth. Consider Fig 2.9, where the growth rates V are plotted against the imposed temperature gradient G. Three different growth zones are visible: equiaxed dendrites, columnar dendrites and planar front. Planar front growth is necessary for single-crystal solidification but is out of the scope of the current discussion due to prohibitive costs and geometric limitations. The red curve represents the Columnar to Equiaxed Transition (CET), as exemplified in 2.8 for casting, which depends highly on the material and can only be considered a rough estimate. Along the $\dot{T}$-lines, the morphology of the growth changes, whilst normal to these lines the microstructures become fines, whilst retaining the same morphology. We can see that AM is a highly capable solidification process, which can produce markedly fined microstructures than casting and can cover all three growth zones given appropriate process parameters.

Rapid solidification processes are typically seen in applications such as AM, welding and powder fabrication. They all have in common that the solidification front grows at high velocities $V > 0.1cm/s$, invalidating many classical analytical solidification models that are based on diffusive equilibrium or interface equilibrium, which cannot be achieved when the diffusion is considerably slower than the solidification front. KURZ et al. (2023) have detailed derivations of analytical models and empirical relationships valid for high cooling rates as well. One result they derive in this way is that the final average grain size depends more on the rate of nucleation of new grains than it does on the growth kinetics, posing high demands for accurate nucleation modelling in any attempt at simulating the microstructure.

The prototypical microstructure in AM, is one dominated by large columnar grains growing anti-parallel to the heat flow directionin the BD of the built part (ZINOVIEVA et al. 2023). While this might seem counterintuitive, as the high cooling rates should lead to fine microstructures, the interaction between subsequent melt lines significantly impacts the overall microstructure, beyond what can be estimated for rapid solidification seen in applications such as welding. Consider Fig. 2.10, where the solidification directions of melt pools nearby can either oppose or amplify one another. This causes clear grain boundaries between ad-

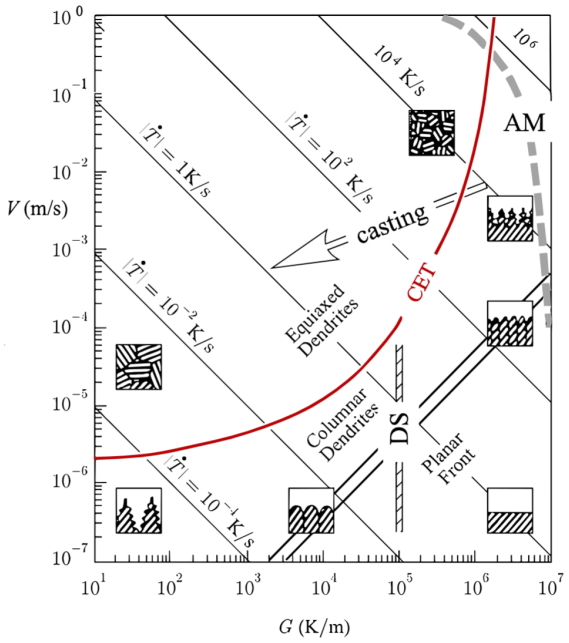


Figure 2.9: Overview of the different solidification regions possible to achieve with different methods. Adapted from KURZ et al. (2023)

jacent scan lines, whilst an unhindered heat flow opposite to the build direction leads to coarse columnar grains growing across several build layers. (BRENNE and NIENDORF 2019). MOSTAFA et al. (2017) observed these columnar structures as well, in the X-Z and Y-Z plane, with Z as the build direction. However, for the X-Y plane a multi-pass behavior was also observed, where grains were larger than in areas that were only molten once.

The rate at which subsequent melt pools influence each other is highly dependent on VED, the volumetric energy density, detailed in Eq. 2.1. MOUSSAOUI et al. (2018) found that a higher VED leads to longer and tighter dendrites oriented along the BD, attributed this to the higher temperature gradients in the BD allowing stronger columnar growth. While VED may correlate well with some microstructural properties, SCIPIONI BERTOLI et al. (2017) compared a constant VED, varying scan speeds and laser power, and found clear upper limits on how high scan speeds could be compensated with increasing laser power before causing irregular solidification phenomena due to complex fluid phenomena in the melt pool. A short overview of methods to simulate both the solidification and thermo-fluiddynamics are outlined in the subsequent sections.

2.4 Simulating Microstructures

Determining any analytical results abound the microstructure in metals is notoriously difficult, in AM it is almost impossible. Figure 2.9 showed the wide range of possible solidification front velocities and temperature gradients that are possible to achieve, with drastically different microstructures. However, several models can approximate the behaviour of crystal nucleation and growth in local scope, enabling physically based simulations to determine the microstructures. The temperature distribution at all times can be assumed as given and will be further discussed in Section 2.5.

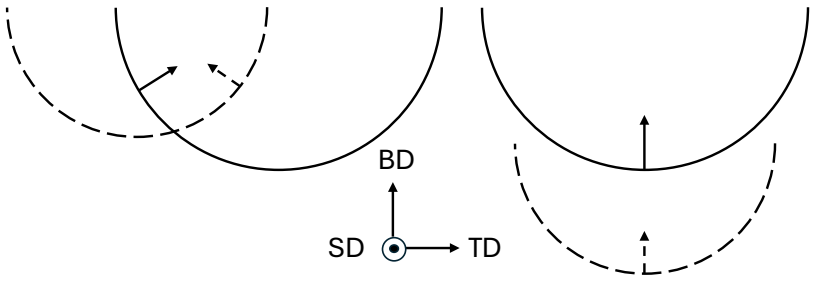


Figure 2.10: Subsequent meltpools, where the dotted lines signify the remains of the previous meltpool. a) Shows the melt pools for two scan lines adjacent to each other, where the solidification directions are opposing. b) Shows the meltpools in subsequent layers, where the solidification directions are parallel. SD: Scan Direction, TD: Traversal Direction, BD: Build Direction. Adapted from BRENNE and NIENDORF 2019

2.4.1 Grain Nucleation

The first key element in the microstructure simulation is the nucleation of new crystals. The strong dependence on the concentration on solid substrates in the melt, the difference in wetting angles θ_c and the temperature leads to great difficulties in determining a nucleation rate by the Eq. 2.20. An often adopted approach is to rather phenomenologically approximate the bulk nucleation rate, where a nucleation rate function $n(\Delta T)$ is assumed to follow a Gaussian distribution. ΔT_c is the critical undercooling for nucleation, and ΔT is the actual undercooling.

$$\frac{dN(\Delta T)}{d(\Delta T)} = \frac{N_0}{\sqrt{2\pi}}\left[-\frac{(\Delta T - \Delta T_c)^2}{\Delta T_\sigma}\right] \tag{2.22}$$

$$N_{discrete}(t) = \frac{N_0}{\Delta T_\sigma \cdot \sqrt{2\pi}} \int_{\Delta T(t-1)}^{\Delta T(t)} \exp\left(-\frac{\Delta T - \Delta T_c}{\sqrt{2}\cdot \Delta T_\sigma}\right) \tag{2.23}$$

The parameters N_0 and ΔT_c must be estimated based on literature, and verified experimentally for each scenario. (KÖRNER et al. 2020). This reduces the predictive abilities of microstructure simulations greatly, and is an active research field. PANZER, BUSS, et al. (2023) found that this phenomenological model is satisfactory if a first-time-right prediction of the nucleation is not needed, but that the additional effort to determine the material parameters for using Eq. 2.20 with Eq. 2.19 is rewarded by significantly improved first-time-right predictions, needed when calibration specifically for the used material and process parameters is not possible.

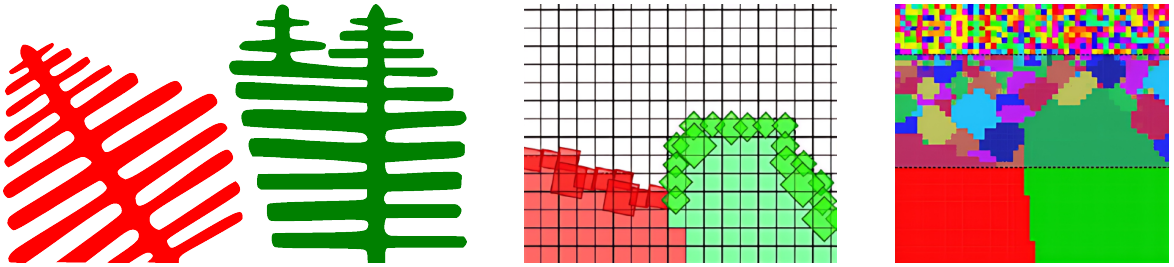
2.4.2 Grain Growth

For the subsequent grain growth there are three different approaches to microstructure modelling that find wide use in literature today; Phase-Field Model, Potts Monte Carlo Model, and Cellular Automata.

Phase-Field Models

The most physical approach is Phase Field (PF), originally developed to model solidification, is now commonly used for all kinds of microstructure modelling. The approach is based on modelling the interface between two phases with a smooth function called a phase-field. It directly uses thermodynamic and kinetic equations with a strong theoretical basis, combined with computational phase diagrams (CALPHAD) for alloyed metals. Most methods based on this have a system of two Partial Differential Equations (PDEs), modelling the evolution of some field describing the concentration or saturation, and an energy field. This allows for very detailed descriptions of the dendrite shapes, as seen in Fig. 2.11. The details of these PDEs may vary, and a discussion and comparison of these is outside of the scope of this thesis, a detailed derivation and discussion is found in KURZ et al. (2023).

Whilst the appeal of these methods is the high resolution of the resulting microstructure, the computational cost is prohibitive for mesoscale modelling. WANG et al. (2019) modelled AM microstructure using a PF method, and needed a mesh spacing Δx at least 1 order of magnitude smaller than the dendrite tip radius, in the sub-μm scale, with a corresponding Δt maximum time step in the 10-20 μs scale following a variant of the Courant–Friedrichs–Lewy (CFL) condition. As solving systems of PDEs on such a scale is so expensive, PF methods are almost unusable in the mm-scale, and even limiting the simulation to 2D WANG et al. (2019) only considered an $80\mu m$ x $80\mu m$ domain. Beyond the computational cost, a very good knowledge of the temperature-dependent material properties is necessary to accurately solve the equations, and there are several calibration parameters that must be fitted to experimental results.



(a) Phase-Field Models **(b)** Cellular Automata with Decentered Squares **(c)** Potts Monte-Carlo

Figure 2.11: A scenario of two adjacent grains growing upwards. Adapted from KÖRNER et al. (2020)

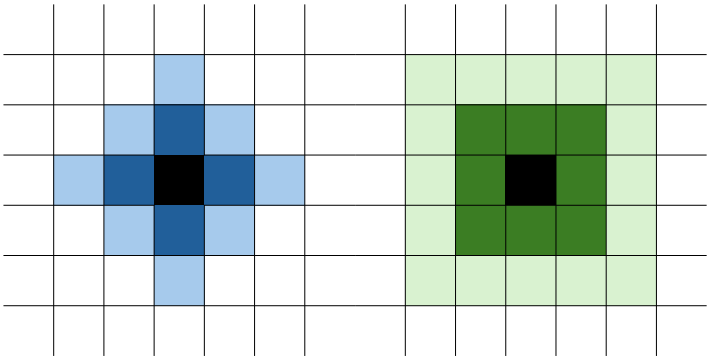
Potts Monte-Carlo Model

Another model is the Potts Monte-Carlo model, which takes a very high-level modelling approach. Here calculations are only performed on the heat-affected zone around the melt pool, the domain divided into a lattice, and all lattices assigned a spin signalling the belonging to a certain grain. An abstract curvature energy of the interface between different grains drives a coarsening, as the system wants to approach a lower energy state. This is implemented by probabilistic transition rules for the spin. An advantage of this abstract and probabilistic model is the greatly reduced computational cost, however, it has not yet been shown to accurately model the competitive grain growth. While useful for other microstructural phenomena, this means that the most important microstructural feature in AM, the columnar grains along the build direction, is not represented using this model. (KÖRNER et al. 2020)

Cellular Automata

To both represent some of the main physical elements of the process, but reduce the computational cost when a detailed dendrite geometry is not needed, a CA based model was devised by RAPPAZ and GANDIN (1993). This method tracks the solid-liquid interface using analytical approximations for the dendrite tip velocity, but without explicitly determining the shape of dendrites. They observed that most dendrite tips have a tightly fitting square envelope, motivating the development of the CA algorithms. Some concepts were borrowed from early Monte-Carlo-based models, such as the bulk nucleation model and the initialization of new grains with random orientations and locations amongst candidate cells. While the first versions had limited usefulness GANDIN and RAPPAZ (1997) developed a model valid for non-uniform temperature fields, and introduced the decentered square and decentered octahedron models for 2D and 3D simulations, respectively. For simplicity, the overarching concept will be outlined here for the 2D case, as well as a short introduction to the main concepts of CA, whilst the implemented 3D case is detailed in 3.1.2.

A cellular automaton is a very simple representation of a complex system, with a few distinguishing features: space is discretized into a lattice of equivalent cells, each cell can have a finite number of discrete states, all state changes depend on a local neighborhood of each cell, and all changes occur simultaneously for all cells at discrete time steps. This simplicity and homogeneity allow for very efficient parallel computation of complex phenomena. There are typically two different types of neighborhoods defined for CA: Neumann and Moore neighborhoods. A Neumann neighborhood of size n contains all cells reachable in n steps, meaning a L_1-distance of n away from the cell center, whilst a Moore neighborhood of size n contains all cells within a L_∞-distance of n away from the cell center, meaning within a cell-centered square of side length $2n$, illustrated for $n = \{1, 2\}$ in Fig. 2.12. For $n = 1$, typically used in CA, this corresponds to the directly adjacent cells belonging to the Neumann neighborhood while the Moore neighborhood also encompasses the diagonal cells. (ILACHINSKI 2001)



(a) Neumann Neigborhoods **(b)** Moore Neighborhoods

Figure 2.12: Neighborhoods of a CA cell. The cell in black is the center, whilst the dark-shaded portions are within the 1st neighborhood, and the lightly-shaded portions are within the 2nd neighborhood.

RAPPAZ and GANDIN (1993) found three states necessary to model microstructures using CA: solid, mushy and liquid, henceforth referred to as states $\xi = (1, 2, 3)$. The solid regime is valid where a cell is cool enough that its grain structure can be assumed as static, while the liquid regime is used for cells hot enough that no grains can exist. All grain nucleation and growth is limited to being modeled within the mushy state. While originally used with fixed transition rules, making the evolution of a CA system entirely dependent on its initial state, this can easily be augmented with probabilistic or physically informed transition rules. For microstructure modeling, a few key transition rules have been found necessary, the first of

which is a temperature-dependent transition rule Ξ_{temp}:

$$\Xi_{\text{temp}} \; : \; \xi_{\text{cell}}^{[t]} = \begin{cases} 3, & \text{if } T \geq T_{\text{liq}}\,, \\ 2, & \text{if } T_{\text{sol}} < T < T_{\text{liq}} \text{ and } \xi_{\text{cell}}^{[t-1]} \neq 1, \\ \xi_{\text{cell}}^{[t-1]} & \text{otherwise.} \end{cases} \tag{2.24}$$

where $x^{[t]}$ refers to any quantity x at the discrete time step t.

Nucleation can be modeled as a probabilistic transition rule Ξ_{nuc},

$$\Xi_{\text{nuc}} \; : \; \xi_{\text{cell}}^{[t]} = \begin{cases} 1 & \text{if } \xi_{\text{cell}}^{[t-1]} = 2 \; \wedge \; RV \; < \; [N(\Delta T)]_{t-1}^{t}\,, \\ \xi_{\text{cell}}^{[t-1]} & \text{otherwise.} \end{cases} \tag{2.25}$$

where RV is random variable in $[0, 1]$, the probability $P(\Delta T)$ is given by the Eq. 2.22, and ΔT is the local undercooling $T_{\text{liq}} - T$. Along with the state change, a new grain is initialized with a random misorientation angle ϕ w.r.t the global coordinate system.

Beyond temperature dependence and nucleation, grain growth must also be modeled. This is the core of CA algorithm and is where the Decentered Square algorithm is employed. Each cell has two additional properties: the center of the decentered square and the length of its sides. The length of its sides evolves as follows:

$$L_{\text{cell}}^{[t]} = \begin{cases} L_{\text{cell}}^{[t-1]} + \Delta t \; V(\Delta T) & \text{if } \xi_{\text{cell}}^{[t]} = 1 \\ 0 & \text{otherwise.} \end{cases}\,, \tag{2.26}$$

where $V(\Delta T)$ is the dendrite tip velocity as a function of the local undercooling, defined according to Eq. 2.21, and Δt is the time passed since the last step.

For the growth itself, any time a mushy cell center is surrounded by a square, meaning the distance d from the cell center to the edge of a square is less than 0, a new square with the same misorientation angle is initialized with its center between the edge of the existing square and the cell center. As seen in Fig. 2.13, this leads to solid cells developing squares that emulate the envelope of the dendrite tips. The details of where a new square center is initialized and on calculating the distance between the cell center and any decentered square is neglected for the 2D case but is outlined in detail for 3D in section 3.1.2.

$$\Xi_{\text{growth}} \; : \; \xi_{\text{cell}}^{[t]} = \begin{cases} 1 & \text{if } \xi_{\text{cell}}^{[t-1]} = 2 \; \wedge \; \exists \; d(L_i) < 0 \text{ for } i \in \mathcal{D}_n, \\ \xi_{\text{cell}}^{[t-1]} & \text{otherwise.} \end{cases} \tag{2.27}$$

where $\mathcal{D}_n$ is the set of solid neighboring cells within a chosen n-th neighborhood, and d is the distance to the square envelope of that cell.

This is by no means a purely CA approach, as it incorporates values dependent on geometries apart from the CA grid, and allows for varying time steps to influence the system's behavior. However, it has proven useful for many microstructural simulations where the grain structure is the main interest. (KOEPF et al. 2018) (MCELFRESH et al. 2024) Contrary to PF models, these methods rely on an external calculation of the temperature field to inform the transitions $\Xi_{\text{temp}}, \Xi_{\text{nuc}}$ and the growth of the square envelopes. Procedures to determine this are outlined in the following section.

2.5 Thermal Analysis

To perform any simulation of the microstructure using CA we need a valid temperature field for our whole simulation domain in both space and time. In this section, we will consider

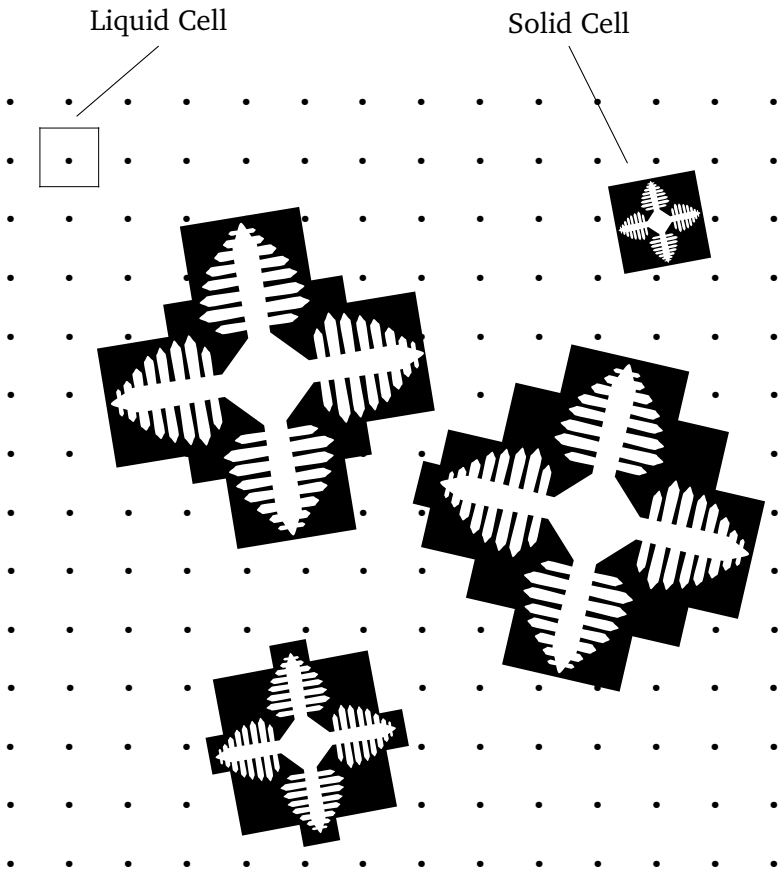


Figure 2.13: Decentered squares on top of the CA grid. Notice that only the squares in the middle of each grain, representing the nucleus, are centered at a CA cell center and that all squares have a rotation corresponding to the grain they belong to. Adapted from RAPPAZ and GANDIN (1993)

several different modeling approaches to this and give a brief introduction to the complete modeling and solving approach used in this thesis. The first step in determining a temperature field is categorizing within the framework of thermodynamics: PBF-LB/M is an open, multi-phase system exchanging energy and mass. With every new layer, new metal powder is added to the system, and all laser impingement on the powder adds new energy to the system, which is transformed into thermal energy upon absorption. As explained in Section 2.1, the powder is packed densely but still has considerable air content and typically achieves a packing density of under 60% of the continuous solid, with an imperfect interface to the solid below it. Once molten, it behaves as a fluid in a small melt pool, before solidifying to become part of a continuous solid with considerable mass. There are a number of choices to make with regard to computation effort, type and quality of desired results, and the amount of calibration necessary. Choices must be made for modeling the laser beam, how this energy is absorbed as heat in the material, and for the heat flow through the body and out of the system.

2.5.1 Heat Source Modelling

When the laser beam hits the powder, a naive approach would be to model the laser beam as a Gaussian-distributed heat flux that is absorbed through the surface. This approach, however, neglects the particle nature of the powder, where the laser beam splits and scatters through the layer. This process can be modeled using ray-tracing, where each ray of a discretized laser is tracked through the powder particles, or a continuum radiation transfer model modeling a continuously distributed laser power and taking into account the packing density of the powder layer. Whilst more accurate, these methods still incur significant computational cost and complexity, so heat sources are still widely used, especially if neglecting fluid phenomena in the melt pool, as this reduces the accuracy requirements for the heat source distribution. (SOUNDARARAJAN et al. 2021) An improvement on the surface heat sources is volumetric heat sources, where the scattering is approximated by a distribution extending below the surface. One of these heat sources is the double-ellipsoidal Goldak heat source by GOLDAK et al. (1984). This model originally developed for welding was adapted from a Gaussian distributed heat flux through the surface, but accounts for the depth the heat can dig into when welding (or here melting) with laser beams. As seen in Fig. 2.14 it is composed of two ellipsoids allowing control of the temperature gradient in the front and the rear of the laser beam independently. This is necessary as heat conduction through powder leads to a steep temperature gradient, whilst heat convection through liquid in the melt pool behind the laser beam is significantly higher, resulting in a gentler temperature gradient.

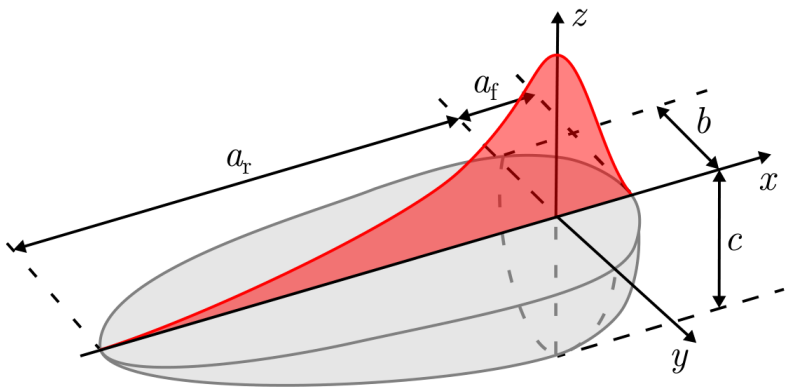


Figure 2.14: The Goldak double-ellipsoidal heat source, here seen for a laser moving towards the right. From BEUERLEIN (2023)

2.5.2 Numerical Model for Temperature Analysis

Once heat flow into the system is determined, the temperature flow throughout the build must be analyzed. To determine the temperature field at every layer, four methods were outlined in SOUNDARARAJAN et al. (2021): FEM, Computational Fluid Dynamics (CFD), Lattice-Boltzmann Method (LBM) and Smoothed-Particle Hydrodynamics (SPH). Of these, the first two use some continuum approximation of the powder, whilst the latter take the particle nature of the powder directly into account. These are also listed in order of increasing computational complexity for a similar resolution of the temperature field but feature additional capabilities. All models but FEM can consider both the particulate nature of the powder layer and fluid flow phenomena within the melt pool. Whilst the FEM and CFD methods build upon

well-established numerical methods, LBM and SPH are still not well enough understood to avoid numerical instabilities, requiring significantly more calibration during simulation. Neglecting the formation of defects and very large deviations from the intended geometry, FEM offers the easiest and cheapest approach for modeling the temperature field. An additional benefit is the capability of FEM for mechanical analysis, enabling coupled thermo-mechanical analysis. As the goal of this thesis is to estimate macro parameters of the material, the FEM approach is detailed more in the following, based on REDDY and GARTLING (2010).

For thermal analysis within solids, no equation is more essential than Fourier's law of conduction in Eq. 2.28, k being the conductive coefficient of the material, $\boldsymbol{q}$ the heat flow and $T(x,y,z)$ the temperature field.

$$\boldsymbol{q} = -k\nabla^2 T \tag{2.28}$$

Using an energy balance equation we can obtain a PDE describing the evolution of the temperature field over space and time.

$$\frac{\partial T}{\partial t} - \nabla \cdot (k\nabla T) = Q \tag{2.29}$$

For a finite elements analysis, these problems are reformulated to a weak form, an integral formulation of the problem where the strong form (Eq. 2.29) is multiplied by a test function v and then integrated over the domain Ω:

$$\begin{aligned} &\int_\Omega v\frac{\partial T}{\partial t}\,\mathrm{d}\Omega - \int_\Omega v\nabla\cdot(k(x,T)\nabla T)\,\mathrm{d}\Omega = \int_\Omega vQ\,\mathrm{d}\Omega \\ = &\int_\Omega v\frac{\partial T}{\partial t}\,\mathrm{d}\Omega + \int_\Omega k(x,T)\nabla v\cdot\nabla T\,\mathrm{d}\Omega = \int_\Omega vQ\,\mathrm{d}\Omega. \end{aligned} \tag{2.30}$$

This is still an analytical form, merely with some assumptions about the domain and boundary conditions. Implemented in a FEM simulation this evaluated numerically for every element in the mesh. This involves reformulating the temperature u and the test function v by basis functions which may differ for each element. Using Galerkin's method, where both the test function and approximation use the same basis function N_i, we obtain:

$$T \approx \sum_{j=1}^{n} U_j N_j\ , \qquad v = N_i \tag{2.31}$$

This is typically solved as in matrix-vector form:

$$M\frac{\mathrm{d}U}{\mathrm{d}t} + K(U)U = F\ , \tag{2.32}$$

that contains a non-linearity due to the temperature dependence of the thermal conductivity. The specific heat capacity c_p is also commonly modeled with a sharp peak at the phase change temperature to incorporate this, leading to a temperature dependence for this as well. The naming of the matrices comes from mechanics, where M is the mass matrix defined as:

$$M_{ij} = \int_\Omega \rho c_p(T) N_i N_j\,\mathrm{d}\Omega\ , \tag{2.33}$$

K the stiffness matrix:

$$K_{ij} = \int_{\Omega} k(x, \sum_{j=1}^{n} U_j N_j) \nabla N_i \cdot \nabla N_j \, \mathrm{d}\Omega \,, \tag{2.34}$$

and F the load matrix, which here contains the volumetric heat source, defined as:

$$F_i = \int_{\Omega} N_i Q \, \mathrm{d}\Omega. \,. \tag{2.35}$$

In addition to the spatial discretization with FEM, a time discretization scheme has to be used as well to analyze the evolution over time necessary in solidification phenomena. Typically, α-one-step finite difference schemes are used, where the value at the next time $[t+1]$ is obtained as a function of the current time derivative, and the time derivative in the next step:

$$T^{[t+1]} = T^{[t]} + \Delta t \left[(1-\alpha) \frac{\mathrm{d}T}{\mathrm{d}t}^{[t]} + \alpha \frac{\mathrm{d}T}{\mathrm{d}t}^{[t+1]} \right], \; 0 \leq \alpha \leq 1 \tag{2.36}$$

It may be observed that this is an implicit set of equations for any $\alpha \neq 0$, which increases computational cost but typically leads to unconditional stability. For $\alpha = 0$, this is the forward Euler scheme requires a CFL-like stability condition.

$$\begin{aligned} \Delta t &< \frac{2}{(1-2\beta)\lambda_{\max}}, \beta < \frac{1}{2} \\ \lambda_{\max} &= argmax_{\lambda} |K - \lambda I| = 0 \end{aligned} \tag{2.37}$$

Chapter 3

Method

As described in section 2.1, the PBF-LB/M is a complex manufacturing process with many components, which is reflected in the complexity of the simulation. The intended workflow of the whole simulation pipeline is outlined in Fig. 3.1. From a software perspective, the modelling and meshing are performed in open-source or commercial CAD and meshing software, the translation of slicing and process parameters to an FEM input by the Python-based Ascent AM Simulations (Ascent), the FEM simulations in FORTRAN-based CalculiX CrunchiX (CCX) and the crystal evolution in a self-implemented Julia-based CA simulation detailed in section 3.1. This thesis aims to add the capability of simulating microstructure evolution to the existing simulation pipeline and devise a model for estimating the fracture criticality based on the microstructure of these regions, the methods of which will be described in this chapter. The employment of a combined CA and FEM has proven itself useful in literature, used successfully for PBF-LB/M simulations by LIAN et al. (2019) and KOEPF et al. (2019).

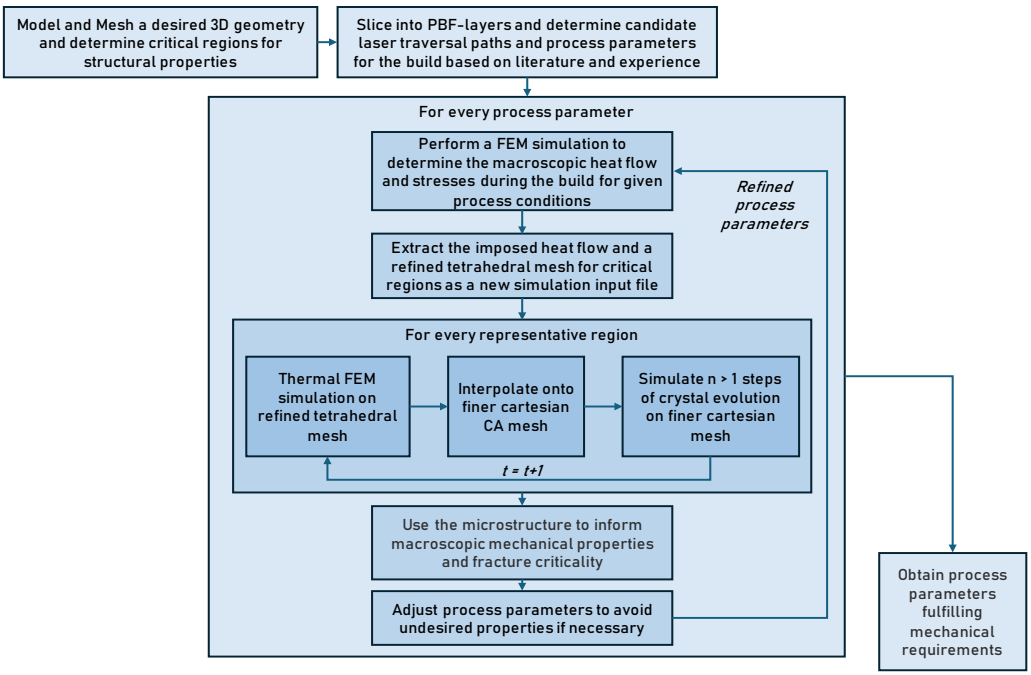


Figure 3.1: Overview of a simulation pipeline within which the microstructure simulation and fracture estimates of this thesis could be integrated.

3.1 Implementation of Microstructure Simulation

The simulation of microstructure evolution has been performed using a custom-built CA crystal simulation code, modelling the nucleation and growth of crystal grains for arbitrary scan strategies. The simulation code works standalone with supplied input files containing the temperature at all nodes of an FEM mesh, and detailing the interpolation between the tetrahedral mesh and the cartesian mesh of the CA. It is also designed to work coupled with CCX, interfacing purely with the C code in a manner compatible with its FORTRAN functions. For coupled simulations, all necessary information is generated at runtime, whilst for standalone simulations, the necessary thermal and geometrical input data can be stored with little additional cost during CCX simulation for later computation of the microstructure.

The implementation can be separated into five function groupings: pre-processing, per-step-processing, crystal growth, additional interfacing for coupled CA-FE simulations running in C, and data exports. The program flow for standalone and coupled runs is detailed in Fig. 3.2 for a single time step. Details of the functionality of each function are outlined in Appendix A.1.

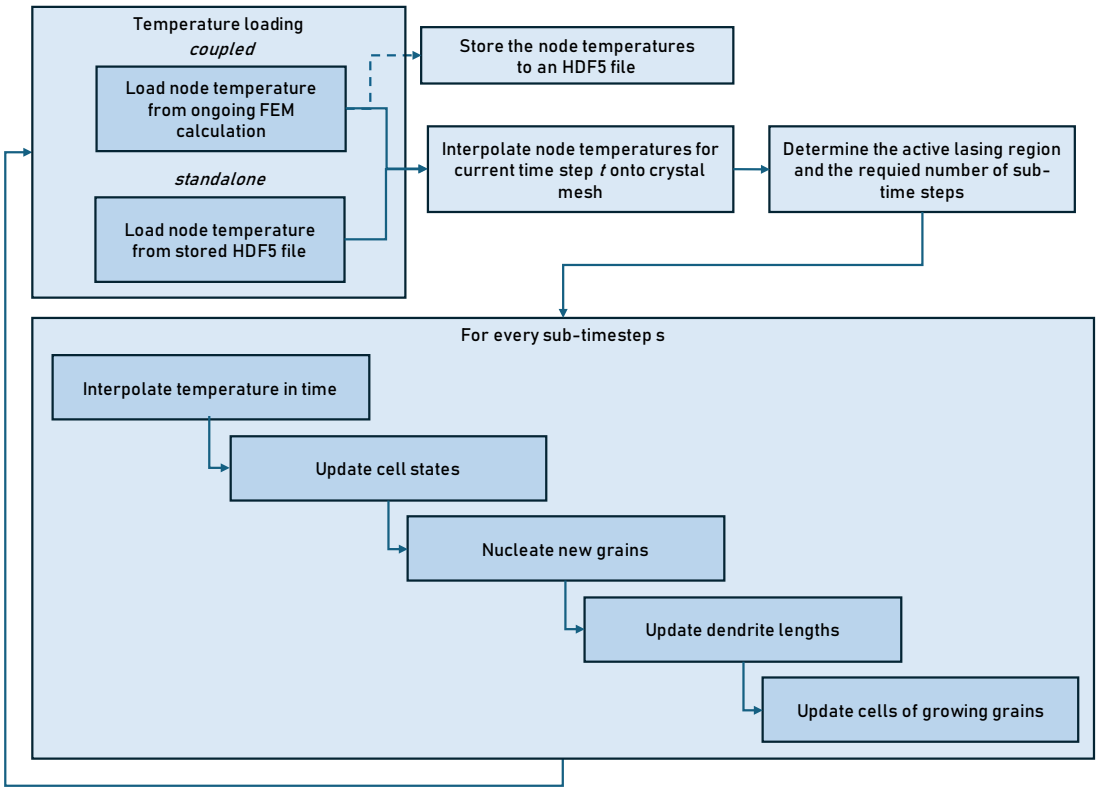


Figure 3.2: The process flow for simulating the microstructure that is performed for every new time-step for which a temperature field is received from the FEM simulation.

3.1.1 Crystal Growth

The CA algorithm employed to simulate the is described in detail in the following sections based on the procedure outlined in RAPPAZ and GANDIN (1993). The data structure associated with the functions is described in section 3.1.3.

Table 3.1: Simulation Parameters used for the multi-layer simulations, with the calibration procedure detailed in Sections 3.1.5 and 3.1.6.

Parameter	Units	Value
Mesh size ΔX	mm	0.002 [1]
Mean undercooling at nucleation ΔT_c	K	12
Standard deviation of undercooling at nucleationΔT_σ	K	6
Maximum nucleation rate N_0	-	3.5e6
Maximum undercooling per CA timestep $\Delta T_{qc,max}$	K	2.4
Powder layer height	mm	0.04
N_x	-	300
N_y	-	300
N_z	-	250

Updating Cell States

Every cell has an assigned cell state ξ_{cell} at every time,

$$\xi_{\text{cell}} = \begin{cases} 0, & \text{if cell is inactive,} \\ 1, & \text{if cell is solid active} \\ 2, & \text{if cell is mushy,} \\ 3, & \text{if cell is liquid.} \end{cases}, \tag{3.1}$$

deciding which of the roles in the algorithm it can perform. Initially all cells are inactive, only changing state once heated above the solidus temperature T_{sol} according to Ξ_{temp}:

$$\Xi_{\text{temp}} : \xi_{\text{cell}}^{[t]} = \begin{cases} 3, & \text{if } T \geq T_{\text{liq}}\,, \\ 2, & \text{if } T_{\text{sol}} < T < T_{\text{liq}} \text{ and } \xi_{\text{cell}}^{[t-1]} \neq 1, \\ \xi_{\text{cell}}^{[t-1]} & \text{otherwise.} \end{cases} . \tag{3.2}$$

Cells can turn mushy and liquid purely based on temperature, whilst the solidification transitions to an active solid are only possible by nucleation or grain growth.

Nucleation of New Grains

All mushy cells have a probability of turning into the nucleus of a new grain given by Eq. 2.22, where all the parameters in the equation are subject to calibration. The integral is calculated for all mushy cells using the QuadGK module for Julia evaluating the integral 3.3, adjusting for the volume of a CA cell. As mentioned in RAPPAZ and GANDIN (1993), nucleation rates N over 1 are problematic, and another calibration parameter $\Delta T_{qc,max}$ determines time step sizes small enough that the nucleation never goes above some chosen value per time step.

$$N = N_{\text{discrete}}(t)\ \Delta x^3 = \frac{N_0 \Delta x^3}{\Delta T_\sigma \cdot \sqrt{2\pi}} \int_{\Delta T(t-1)}^{\Delta T(t)} \exp\left(-\frac{\Delta T - \Delta T_{\text{c}}}{\sqrt{2} \cdot \Delta T_\sigma}\right) \tag{3.3}$$

Once a cell is nucleated, its state is set to solid active ($\xi_{\text{cell}}(t) = 1$), and a new grain is initialized with random crystallographic orientations between -44° and +45° as suggested in RAPPAZ and GANDIN (1993) due to a four-fold symmetry in the crystallographic space. Its octahedral lengths are initialized to zero, as a freshly nucleated nucleus would have a size smaller than the spacing of the CA cells (KURZ et al. 2023). Its octahedral center is initialized to the Cartesian coordinates of the cell center.

Interface Transitions

Interface transitions are an optional probabilistic state transition when simulating the melting of an existing solid metal piece, where a possible state transition exists for mushy cells adjacent to an existing inactive solid cell, allowing existing grains in the metal to grow as well. The transition rule Ξ_{int} is defined as follows:

$$\Xi_{\text{int}} : \xi_{\text{cell}}^{[t]} = \begin{cases} 1 & \text{if } \xi_{\text{cell}}^{[t-1]} = 2 \ \wedge \ \xi_{\text{N1}}^{[t-1]} = 0 \ \wedge \ RV \ > \ P_{\text{threshold}} \\ \xi_{\text{cell}}^{[t-1]} & \text{otherwise.} \end{cases} , \tag{3.4}$$

where N1 is the set of neighbors in the 1st Neumann neighborhood, and $P_{\text{threshold}}$ a calibration parameter. The transition rule is applied for all neighbors, meaning a cell surrounded by solid cells has a $(1-P_{\text{threshold}})^6$ probability of turning solid at every time-step. $P_{\text{threshold}}$ is subject to careful calibration and gives a probability of a mushy cell not becoming part of a neighboring solid grain. $P_{\text{threshold}}$ below 0.95 typically dominated over nucleation, even at higher nucleation rates than used in the final simulations. This additional calibration is avoided in newly built parts, where this transition is unnecessary as all grains within the simulation domain are modelled starting from nucleation.

Dendrite Growth

Due to practicalities step this is performed separate from the octahedral growth algorithm. Here the dendrite lengths are updated, represented by the distance from the octahedral cell center belonging to a cell, to its faces. This growth calculated for all active solid cells ($\xi_{\text{cell}}(t) = 1$), and is approximated according to Eq. 3.5, where v is defined according to Eq. 2.21.

$$L^{[t]} = L^{[t-1]} + \Delta t_{\text{loc}} v(\Delta T^{[t]}) , \tag{3.5}$$

where the Δt_{loc} is the local time-step size.

3.1.2 Updating Cells of Growing Grains

The core of CA-based microstructure simulations is capturing physical grain growth. All mushy cells ($\xi_{\text{cell}}(t) = 2$) have the capability of belonging to a grain, but do not yet. These take the role of victims of engulfing if their cell center is within the octahedral dendrite envelope of another grain. The active solid cells ($\xi_{\text{cell}}(t) = 1$) belong to a grain, and take the role of capturing cells. An important note for the implementation is that iteration over the mushy cells is drastically easier computationally, as the number of mushy cells is typically fairly small, and iterating away from the cell in increasing distance guarantees that any capture calculated is also valid. If iterating outward from the existing active cells, as may be intuitive, there is a significant risk of calculating captures of cells also engulfed by other cells that are closer. The code has been implemented to allow for iterations to cells in the n-th Moore or Neumann neighborhood for the cell, where larger neighborhoods do not increase the computational cost much due to iterating outwards, but there is still an associated overhead. The 2nd Moore neighborhood (which includes 124 cells) has been chosen as a compromise, but for most cases, a 1st Moore neighborhood should provide the same results as almost all captured cells are within this neighborhood. The order of iterations is randomized within each

neighborhood to avoid a bias in any direction but is still retained to an iteration order 1st Neumann, 1st Moore, 2nd Neumann, etc, such that the physically closest cells are checked first. The procedure of calculating whether an octahedron has engulfed a neighboring cell center and determining the center and size of the octahedron of the newly captured cell is outlined below.

Decentered Octahedron Algorithm

Intuition

GANDIN et al. (1996) observed that most dendrite tips grow uniformly out from a main branch. For most metals, it is possible to fit a rectangular envelope rather closely around this dendrite tip, motivating the modeling of dendrite envelope growth rather than the detailed geometry of the dendrites themselves. In GANDIN and RAPPAZ (1997) this concept was extended to 3D, with a decentered regular octahedron rather than previous 2D attempts using rectangles or centered squares with compensating factors. It is decentered in that the center of the octahedron moves along with the tip of the dendrite, only centered at the coordinate points of the cell it belongs to if it is the nucleus of the grain.

Determining Distances

To determine whether a capture event should occur, it has to be determined whether the octahedron has grown enough to engulf the center of a potential victim mushy cell. This is determined separately 8 times for each face of a capturing active solid cell, shown exemplarily for a cell that is not captured in Fig. 3.3. The first step is obtaining the surface normal vectors of every cell in the global coordinate system. As each grain has a spacial orientation described by ϕ_1, ϕ_2, ϕ_3 relative to the global system, this can be obtained by transforming local face normal vectors by a grain-specific rotation matrix. As these are Euler angles, an Euler rotation matrices are composed to rotate the vector, composed of three matrices rotating about the Z, X, and then the new Z axis, each rotating the system by ϕ_i.

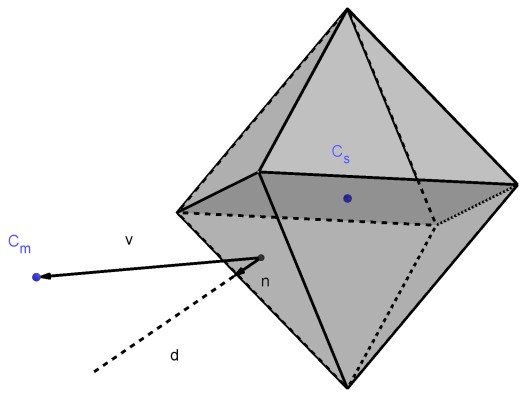


Figure 3.3: Distance calculation for a cell being captured by an octahedron.

$$R = R_z(\phi_3)\, R_x(\phi_2)\, R_z(\phi_1) \tag{3.6}$$

where the rotation matrices R_z and R_x are defined as:

$$R_z(\phi) = \begin{bmatrix} \cos(\phi) & -\sin(\phi) & 0 \\ \sin(\phi) & \cos(\phi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix} . \tag{3.7}$$

The surface normal vector for face f is then defined as:

$$\boldsymbol{n}_f = R(\phi_1, \phi 2, \phi_3)\boldsymbol{n}_{f,loc} , \tag{3.8}$$

where $\boldsymbol{n}_{f,loc}$ is the normal vector of face f in the local coordinate system, which is constant across all regular octahedrons.

Given the coordinates of the octahedral center of the capturing cell c_s, the side lengths of the octahedron l_s, the cell center of the mushy cell c_m, it is straightforward to determine the distance, giving a simple criteria $d < 0$ for capturing. d is defined as

$$d = \boldsymbol{v} \cdot \boldsymbol{n}_f , \tag{3.9}$$

where

$$\boldsymbol{v} = c_m - c_{s,f} . \tag{3.10}$$

$c_{s,f}$ is the centre of the face f of the solid cell, defined as:

$$c_{s,f} = c_s + l_s \boldsymbol{n}_f . \tag{3.11}$$

Initializing a New Octahedral Cell

Once a capturing solid cell and a victim mushy cell have been found, the octahedral center of the newly captured cell has to be determined, an illustration of the locations determined in the intermediate steps is seen in Fig. 3.4

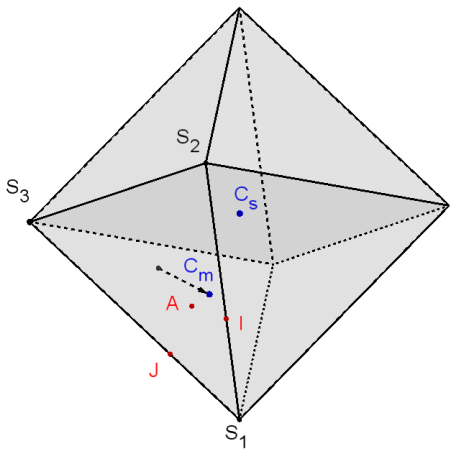


Figure 3.4: Intermediate calculations distances to determine the location and size of a new octahedron.

1. Determine the projection A of the vector c_s to c_m onto the capturing face f

2. Determine the corner S_1 of f, which is closest to A. This can be done by rounding the entires of the surface normal vector to the closest integer in its local coordinate system, returning the corners of the face in the local coordinate system.

3. Project A onto the edges S_{12} and S_{13} to obtain the points I and J respectively.

4. Find the lengths IS_1, IS_2, JS_1, JS_3

5. Let the intial size L_m of the new octahedron be given by:

$$\begin{aligned} L_m &= \sqrt{\frac{2}{3}} max(L_{12}, L_{13}) \,, \\ L_{12} &= \frac{1}{2}\left(min\left(IS_1, \frac{\Delta x}{\sqrt{3}}\right) + min\left(IS_2, \frac{\Delta x}{\sqrt{3}}\right)\right) \\ L_{12} &= \frac{1}{2}\left(min\left(JS_1, \frac{\Delta x}{\sqrt{3}}\right) + min\left(JS_3, \frac{\Delta x}{\sqrt{3}}\right)\right) \,. \end{aligned} \tag{3.12}$$

   This corresponds to truncating the new octahedron to retain the same outer boundary as the capturing octahedron.

6. Let cell center $c_{m,oct}$ be:

$$c_{m,oct} = S_1 + \sqrt{3} L_m \frac{c_s - S_1}{\|c_s - S_1\|} \tag{3.13}$$

3.1.3 Data Structure

The data structure of the arrays is described in Table 3.2, where nz can be replaced by currNz for any multi-layer run with a reduced region of interest.

Table 3.2: Overview of the Arrays used within the CA simulation.

Name	Description	Dimensions	Datatype
cellIDArray	Stores the ID of the grain that each CA cell belongs to.	[nx,ny,nz]	Int32
cellStateArray	Stores the state of each CA cell.	[nx, ny, nz]	Int8
cellOctaArray	Stores the proxy grain center according to decentered octahedron algorithm and the side length of the octahedron.	[nx,ny,nz,4]	Float64
grainArray	Stores the angles of each grain, to be used in the decentered octahedron algorithm. Generated randomly between -44 and +45 at nucleation. The column index corresponds to the grain ID.	[3,numGrains]	Float32
temp prevTemp locTemp prevLocTemp	Stores temperature of the current time step Stores temperature of the previous timestep Stores temperature of the current sub-timestep Stores temperature of the previous sub-timestep	[nx,ny,nz]	Float32

3.1.4 Increasing Computational Efficiency

Although orders of magnitudes cheaper than PF simulations, the computational cost of a 3D CA simulation is still significant, especially in multi-layer simulations. KOEPF et al. (2018) spent over 200h of computation time on a problem significantly smaller than what is intended to be simulated here, even with a computation cluster at their disposal. An efficient method for limiting the computational domain and a robust parallelization scheme has been devised to combat this prohibitive computation time. In multi-layer simulations, it is pointless to compute anything for the portions of the domain that do not yet exist, motivating the idea of bounded computational domains in the BD. Additionally, there is very little change in the crystal structure below T_{sol}, giving a lower bound in the BD. To ensure that columnar grain growth can occur, some margin must be preserved to let solid grains grow into the liquid domain of the current build layers. Assuming Z as the BD, the domain is limited by z_{lower} and z_{upper}.

The average temperature $T_{\text{l}}(z)$ at each layer is computed as

$$T_{\text{l}}(z) = \frac{1}{n_x n_y} \sum_x \sum_y T(x,y,z) \,, \tag{3.14}$$

where n_x and n_y are the number of cells in x- and y-direction respectively. The heigh of the current laser impingement $z_{\text{hot}}(t)$ is computed as

$$z_{\text{hot}}(t) = argmax_z T_{\text{l}}(z) \,. \tag{3.15}$$

Adding a margin of safety, the active computation area is bounded in z-direction as:

$$z_{\text{active}} = z \;:\; z_{\text{lower}} \leq z \leq z_{\text{upper}} \tag{3.16}$$

$$z_{\text{upper}} = z_{\text{hot}} + \frac{n_z}{20} \tag{3.17}$$

$$z_{\text{lower}} = z_{\text{hot}} - 2d_M \,, \tag{3.18}$$

where d_M is the depth of the melt pool, the lower bound should be verified visually with larger margins to ensure no remelting occurs at depths $z_{\text{hot}} - 2d_M$.

In addition to reducing the computational domain, a 4 step select-test-resolve-update pattern has been implemented to ensure a robust parallelization, using Julia's robust array indexing system.

1. Select all cells that are possible candidates, storing their indices.
2. Test each cell for a condition, typically iterating over cells in some n-th neighborhood, and store all relevant indices in a separate vector for each thread.
3. Combine all the relevant indices, resolving any duplicates or conflicts.
4. Update all cells

This approach has some noticeable overhead regarding the storing of indices. Still, it has consistently outperformed other approaches at parallelization during this work due to the heterogeneity of the workload, which is hard to estimate before dividing the computational domain between threads. With this approach all threads have a similar workload, improving the scaling of the problem.

3.1.5 Validation and Verification Procedure for Microstructure Simulation

It is essential to validate and verify the output of the microstructure simulation. According to SARGENT (2010) a valid simulation model relies on an iterative process of experimentation, modelling and computer programming, as illustrated in Fig. 3.5. The system investigated is the PBF-LB/M build process, and the CA model, including nucleation and dendrite growth models are given as a conceptual model. To ensure a valid simulation, verification has to be performed to ensure the implementational correctness of the conceptual model, and operational validation comparing experimental results with simulation results. This section will outline the methods employed to ensure a correct physical modelling of the microstructure.

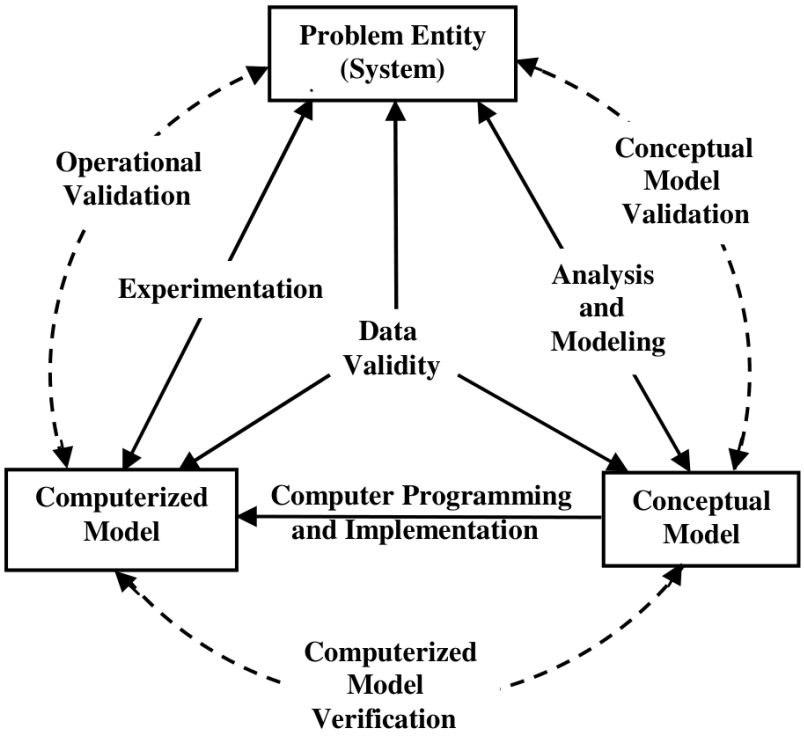


Figure 3.5: Overview of the iterative modelling, verification and validation procedures for physical simulations. (SARGENT 1984)

Verification

Verification is performed by dynamic and static testing to ensure the correctness of the implemented functions. All functions should be tested with a range of sample inputs to ensure the desired input-output relationship, and scenarios should be devised where an analytical solution is available, for example, no nucleation, temperatures colder than T_sol, and simplified heat flows. Additionally, many of the simulation steps rely on linear time-flow, where calculation step b can only be performed once a has been completed. A structured walkthrough of this should also be done, taking special note of parallelized components

Convergence Analysis

A convergence study is performed to find a suitable mesh size for the CA simulation. KOEPF et al. (2018) used a Δx of 10 μm, whilst ANSARI DEZFOLI et al. (2021) used 0.2 μm. Two simple

test cases, the bottom layer of a multi-layer simulation and a single-track melt simulation, will be performed to show convergence towards the Δx used in subsequent simulations. For each scenario, the simulation will be performed for $\Delta x = \{10\mu m,\ 5\mu m,\ 2\mu m,\ 1\mu m\}$. The average volume-weighted aspect ratio of the grains AR_v has been chosen as a convergence metric, defined as:

$$AR_v = \frac{V_{cell}}{V_{tot}} AR \ , \tag{3.19}$$

where AR is defined in eq. (3.26) on page 37 and the calculation procedure is detailed in 3.2. This is done as the number of grains in total may increase further with smaller Δx, as the real number of grains consistently is higher than the estimates according to phenomenological models (KURZ et al. 2023). These grains would be smaller than possible to capture with a given Δx, whilst not influencing the macro-scale properties significantly.

Comparison with Experimental Results

Beyond convergence, a comparison to existing experimental results is necessary for verification. The surrounding geometry and process conditions used in the simulation have been investigated experimentally by ZHAO (2024), where the average surface area per grain increased with increasing laser power, and with representative volumes extracted from higher layers in the geometry, illustrated in Fig. 3.6. This is because higher heat flow leads to larger grains, and the narrowing around region 0, seen in Fig. 3.7 leads to higher temperatures in the upper regions. This should be reflected in the simulation output. The material parameters used for the thermal and microstructure simulation are found in Table 3.3.

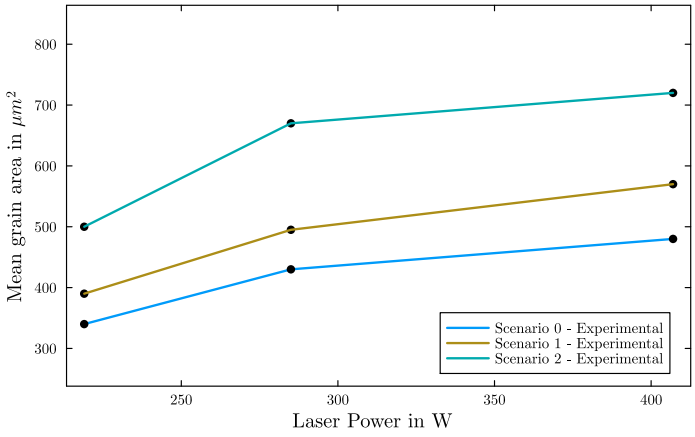


Figure 3.6: Mean grain areas corresponding to regions 0, 1, and 2 in Fig. 3.7. Adapted from ZHAO (2024)

Another qualitative validation is performed with the results of MOSTAFA et al. (2017), which analyzed the grain structure of IN718 produced by PBF-LB/M produced with a VED of 100 kJ/mm^3, close to 100 kJ/mm^3, the VED of a 417W test scenario used in this thesis. They found that the $x-y$ plane showed a clear difference between Zone I, which was only molten once, and Zone II, where the melt pools of adjacent laser tracks overlapped. Zone I showed much smaller grains, at 30 μm average, contrasting to the average of 50 μm for Zone II. This should also be confirmed in the resulting microstructure.

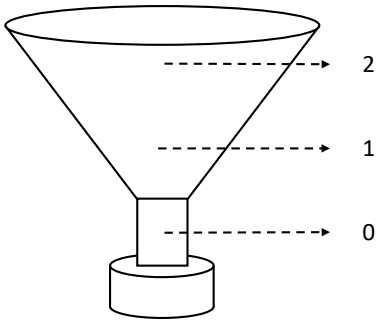


Figure 3.7: Schematic geometry of the experimentally analyzed part, indicating the location of extracted regions that were investigated. Adapted from ZHAO (2024)

Table 3.3: Material constants for IN718

Parameter	Units	Value	Reference
Melting Temperature T_{liq}	K	1609	MILLS (2002)
Solidification Temperature T_{sol}	K	1533	MILLS (2002)
Diffusion coefficient D_L	m^2s^{-1}	1.6e-9	NIE et al. (2014)
Gibbs-Thompson Coefficient Γ	Km	3.65e-7	NIE et al. (2014)
Partitioning Coefficient k_0	-	0.48	NIE et al. (2014)
Initial Concentration C_0	$wt.\%$	0.5	NIE et al. (2014)
Liquidus Slope m_L	$K/wt.\%$	-10.5	NIE et al. (2014)

3.1.6 Calibration of Nucleation Parameters

The above results from MOSTAFA et al. (2017) are visible after a few simulated layers, meaning computation times are relatively short before a structure to calibrate after has become visible. As nucleation has to be separately adjusted for single-layer and multi-layer simulations, nucleation parameters and adaptive time-stepping have been calibrated for the simulation to exhibit behavior as described. With excessive nucleation, the growth inward from the edge of melt pools is prevented by nucleation within the melt pool. When nucleation is too low, no grains form in the middle of the melt pool, and crystals often grow across adjacent melt pools in the same layer. Once a range of nucleation parameters avoiding these two conditions has been found, some sample tests can be performed to determine a curve fitting, with the grain density as a function of N_0, the maximum nucleation number. Based on this function, an appropriate N_0 can be selected based on the desired grain density.

3.2 Extracting Microstructure Features

An abstraction of the microstructure is required for further work on the fracture criticality. Based on the results by VIEILLE et al. (2020) and BRENNE and NIENDORF (2019), the size, orientation, and shape of the grains have been selected as candidates for data to inform fracture parameters, and this section will describe how this information is obtained from an existing crystal mesh.

A simple count of the number of cells belonging to a certain grain will suffice to determine

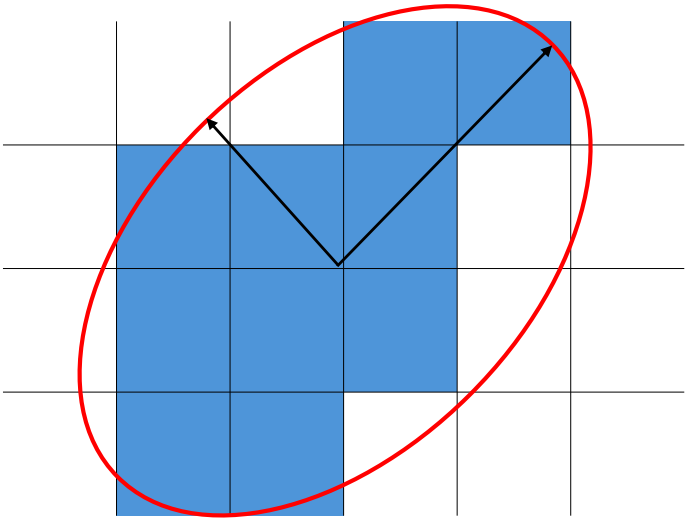


Figure 3.8: Schematic illustration of an ellipsoidal approximation. Adapted from McElfresh et al. (2024)

the volume or area of a grain. The area is obtained when slicing the CellID array along a given x,y, or z coordinate, whilst the full CellID array gives the volumes of all grains. Whilst computationally more costly, determining all statistics for a 3D field is a minor additional cost compared to the simulations performed to obtain the microstructure.

Determining the shape of the grain has been done using ellipsoidal approximations of the grain shape as a 3D extension of the approach used by McElfresh et al. (2024) for crystal plasticity modeling. This approximation is illustrated for a 2D ellipse in Fig. 3.8, where the direction $\boldsymbol{v_i}$ and lengths d_i of the axes define the ellipsoid. The shape and size are described using the gyration tensor,

$$\boldsymbol{R} = \frac{1}{N} \sum_{i \in \mathcal{D}} \tilde{r}_i \otimes \tilde{r}_i \ , \tag{3.20}$$

where the distance from the center of the ellipse is

$$\tilde{r}_i = r_i - c \ , \tag{3.21}$$

and the center is defined as:

$$c = \frac{1}{N} \sum_{i \in \mathcal{D}} \boldsymbol{r_i} \ . \tag{3.22}$$

Here r_i are the coordinates of the cell centers, and $\mathcal{D}$ is the set of all cells belonging to the grain. To avoid outliers, the possible cell centers c_j are computed n times, where

$$c_j = \frac{1}{N-1} \sum_{i \neq j} \boldsymbol{r_i} \ , \tag{3.23}$$

and $c = median(c_j)$. Furthermore, $\tilde{r}_i$ is computed for all points, and the 20 % longest distances are discarded. This comes at a high additional cost but has proven necessary, as the sum of the volume of all ellipsoids, based on the full set of points, far exceeds the total volume of the computed part.

The orientation and shape of the ellipsoid can be obtained using eigendecomposition of $\boldsymbol{R}$:

$$\lambda_i, \boldsymbol{v_i} \ : \ \lambda_i \boldsymbol{R} = \boldsymbol{R} \boldsymbol{v_i} \ , \tag{3.24}$$

giving the axis lengths λ_1, λ_2 and λ_3 in decreasing magnitude, and their respective directions defined by the eigenvectors $\boldsymbol{\nu_1}, \boldsymbol{\nu_2}$ and $\boldsymbol{\nu_3}$. Based on this, we can obtain two important grain characteristics: the grain size D using the radius of gyration,

$$D = \sqrt{\lambda_1^2 + \lambda_2^2 + \lambda_3^2}\,, \tag{3.25}$$

and the aspect ratio

$$AR = \frac{\lambda_3}{\lambda_1}\,. \tag{3.26}$$

Note that some grains may be entirely described within a slice of the CellID array, where the gyration tensor will not have full rank, but we nevertheless want the grain size and aspect ratio. Here λ_3 is set to Δx.

Similarly, all these values can be obtained for 2D ellipses by applying the above procedure to a slice of the CellID array. The 2D values can be calculated as an average over a range of slices to achieve more representative results.

3.3 Microstructure-Aware Fracture Modeling

As established in Section 2.2.3, the microstructure significantly influences the fracture criticality of metals, but the details of how the fracture mechanism is influenced is still not thoroughly explored (BRENNE and NIENDORF 2019). Extensions of available fracture models, or the development of novel models, to account for microstructure is still an active research topic. An extension of the JC fracture model will be presented in this section, along with experiments necessary for calibration, along with a short outlook on a more physical approach that could be implemented in the future.

3.3.1 Derivation of an Extended Fracture Model

PANZER et al. (2024) studied the fracture behavior and grain density of different specimens with varying VED, showing that the fracture toughness increased with increasing mean grain size, as seen in Fig. 3.9. BRENNE and NIENDORF (2019) showed that fractures could be halted by deflecting cracks where large columnar grains, aligning with the BD, dominate the microstructure. This happened at an angle of around 45° to the BD. Both of these observations allow for higher fracture toughness when grain sizes increase, whilst the above formulation of the JC has no way to account for this. However, as the JC fracture model is purely empirical it can be extended with further multiplicative factors to account for the microstructure. The existing JC model defines the strain at failure ε_f as:

$$\varepsilon_e^f = (d_1 + d_2 \exp(d_3 \sigma^*))\left(1 + d_4 \ln \dot{\varepsilon}^*\right)(1 + d_5 T^*)\,, \tag{3.27}$$

To model the dependence on the grain size, a factor is proposed:

$$(1 + d_6 \delta^*)\ , \tag{3.28}$$

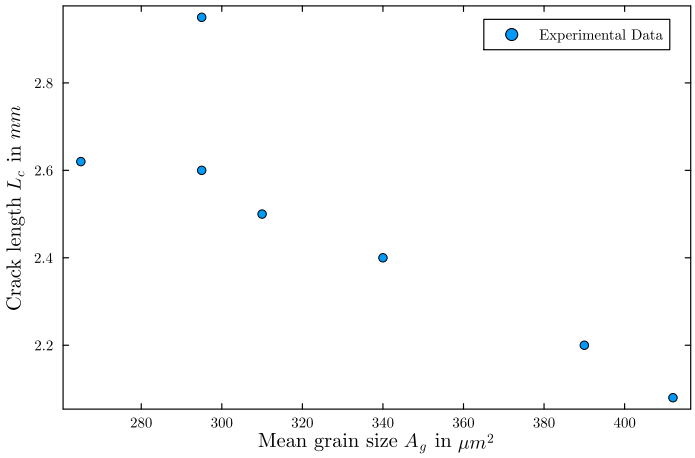


Figure 3.9: Experimental results correlating crack lengths and grain sizes. Based on PANZER et al. (2024).

where d_6 is a calibration parameter and δ^* is the relative mean grain size defined as:

$$\delta^* = \frac{A_\mathrm{g} - A_\mathrm{g,ref}}{A_\mathrm{g,ref}} \tag{3.29}$$

where A_g and $A_\mathrm{g,ref}$ are the mean grain areas of the desired specimen, and of a reference specimen, respectively.

Beyond the grain size, the anisotropy of the fracture criticality will be modeled by another factor:

$$\left(1 + d_7\ \delta^* sin^2(2\vartheta^* + d_8)\right)\ , \tag{3.30}$$

where d_7 and d_8 are calibration parameters. ϑ^* is the angle between the crack direction and the average direction of large columnar grains in the material. This is done as the reasoning in BRENNE and NIENDORF (2019) does not necessarily lead to a dependence on the BD, as the crack deflection takes place by intergranular fracture along these grains boundaries and some processing parameters can lead to columnar grains growing angled in the traversal direction of the laser (PARIMI et al. 2014). Only large grains were seen to have a noticeable impact, so δ^* is included as a factor to account for this. By the method described in Section 3.2, one can determine the aspect ratio AR, volume V, and direction of the major axis $\boldsymbol{\nu_1}$, of the ellipsoid approximation of every grain. A measure for the average direction of elongated grains $\hat{\boldsymbol{\nu}}$ can be obtained by

$$\hat{\boldsymbol{\nu}} = \frac{\tilde{\boldsymbol{\nu}}}{||\tilde{\boldsymbol{\nu}}||}\ , \tag{3.31}$$

$$\tilde{\boldsymbol{\nu}} = \frac{1}{n_g} \sum_{g \in \mathrm{grains}} \boldsymbol{\nu_{1,g}} \left(AR_g - 1\right) V_g\ . \tag{3.32}$$

The factor $\left(AR_g - 1\right)$ ensures that mainly elongated grains contribute to the direction, and the volume weighting ensures that the large columnar grains have more impact. The angle can then be determined as

$$\vartheta^* = arccos\left(\hat{\boldsymbol{\nu}} \cdot \hat{\boldsymbol{f}}\right) \tag{3.33}$$

$\hat{\boldsymbol{f}}$ being the unit vector of the crack direction. For simplicity $\hat{\boldsymbol{\nu}}$ can also be assumed to be the BD if experimental or simulative results for the applied process parameters indicate columnar

grain growth clearly along the BD. The calibration parameter d_7 determines the magnitude of the impact of this crack deflection on the fracture toughness. As 45° is not necessarily exactly the angle at which the deflection is most probable, d_8 allows for adjustments to this. In total, this extended JC model takes the following form:

$$\varepsilon_e^f = (d_1 + d_2 \exp(d_3 \sigma^*)) \left(1 + d_4 \ln \dot{\varepsilon}^*\right) (1 + d_5 T^*) (1 + d_6 \delta^*)(1 + d_7\ \delta^* sin^2(2(\vartheta^* + d_8))) \quad (3.34)$$

3.3.2 Calibration Procedures

As the fracture model is purely empirical, all parameters d_i must be calibrated using fracture experiments. Calibration of parameters d_1 to d_5 is treated in PANZER (2020) and will not be discussed further, but must be performed before calibrating parameters d_6, d_7, and d_8.

Parameter d_6 modeling the magnitude of the influence grain size has on the fracture toughness should be calibrated with at minimum three fracture experiments at differing grain sizes, all with a loading angle perpendicular to the BD. The reference grain size $A_{g,\text{ref}}$ should be chosen as the smallest mean grain size amongst the investigated specimens. Any validity of the fracture model should be limited to grain sizes between the ones used for calibration.

Parameters d_7 and d_8 should be calibrated only once d_6 is determined, with experiments varying the crack direction against specimens with a similar grain structure. First, d_8 is calibrated, slightly varying the angle around 45°, where d_8 can be determined as the offset from 45° demonstrating the highest fracture toughness. Finally, d_7 can be calibrated, by comparing the fracture toughness of a 90° angle, meaning no impact of columnar grain directions, and at an angle $\alpha = 45° + d_8$ where the deflection is the highest.

3.3.3 Outlook on CPFEM Fracture Modeling

Empirical fracture models have seen the addition of CPFEM as a more physics-based approach that explicitly models the anisotropy of the elastic tensor dependent on the grain orientation within a material (ROTERS et al. 2010). LONG et al. (2023) used CPFEM to study the influence of the crystal structure on Fatigue Crack Growth (FCG), limited 2D and manually modeled crystals structures from experiments. SIDDHARTH et al. (2024) studied micro-crack propagation, also using CPFEM, on randomly generated microstructures emulating equiaxial growth zones, also limited to 2D. CPFEM has been used for simulating mechanical properties of Powder Bed Fusion (PBF) as well, but fracture modeling has been neglected (LI et al. 2020), (CHAKRABARTY et al. 2023). As detailed in Section 2.3.3, the microstructures of AM materials have significantly more heterogeneous properties, sometimes with clear boundaries within a single part if process parameters are non-constant as in BRENNE and NIENDORF (2019). As applying CPFEM to model fracture in such heterogeneous microstructures has not been previously attempted, modeling the fracture criticality using CPFEM is outside this thesis's scope.

Chapter 4

Results and Discussion

This chapter will present results of the microstructure simulation and fracture modelling. The verification and validation of the implemented model will be treated in the subsequent sections, performed as described in Section 3.1.5. Subsequently, the obtained results of the fracture model are presented before addressing the limitations posed by the microstructure simulation and the fracture estimates.

4.1 Verification of Microstructure Simulation

Verification contains all comparisons between simulation outputs and what can be deduced directly from the conceptual model analytically. This is performed through static testing by structured walk-throughs and structure analysis, and dynamic testing by checking input-output relations with changed simulation parameters.

4.1.1 Static Testing

Static testing has been performed using the Julia debugger in an interpreted mode, rather than a compiled mode, where the values of all variables and arrays can be investigated at all simulation steps. A non-exhaustive list of possible logical errors investigated is: race-condition in parallelisms, chain reactions during the application of a single CA transition, a mismatch between indexing of different arrays, and violation of linear time-flow.

The CA state transitions depend on the cell state, the temperature, and the octahedron properties. Exploiting this, the select-test-resolve-update mechanism for parallelism described in Section 3.1.4 selects the indices of all candidates for a transition based on their state before a transition rule is applied, and only updates the states of the cell once the other components of the transition have been applied. Using this method, all race-conditions and chain reactions have been avoided in the code. Every thread stores the number of changes applied in each time step, where the sum should correspond to the difference in the total number of cells in the given state. The number of cells changed should at most equal to the number of candidate cells for transition, before the transition rule is applied. Both of these cases have been verified by dynamic testing where the logs were investigated, comparing these values for all time steps.

Indexing correctness have been ensured by type-checking of all components during debugging, and by Julia's broadcast function that necessitates an explicit dot-operator to differentiate whether any function is applied on a scalar, or element-wise on vector or arrays. Timeflow correctness has been ensured by structured categorization of loops and functions into either spatial or temporal iteration, where all parallelism is avoided in temporal iterations.

4.1.2 Dynamic Testing

Dynamic testing of input-output relations has been performed by visualizing the output using The ParaView Visualization Package (ParaView) and data analysis of the simulations logs. Some model and simulation parameters have been adjusted, to gauge the impact on the simulation output.

Verifying nucleation parameters, for any multi-layer simulation with no existing grains, it is clear that there will not be any solid cells if nucleation is disabled. This has been verified by assigning the maximum nucleation rate $N_0 = 0$, where no cells solidified. The opposite is the case when N_0 is set high enough that it will result in $N(\Delta T) > 1$ for any undercooling, where the number of solid cells should correspond to the number of grains, as nucleation probability of any cell experiencing undercooling is 100%. For the single-track scenario, $N_0 = 0$ should correspond to solidification based purely on the growth of pre-existing grains around the melt track, which was verified visually, seen in Fig. 4.1 and by a static number of grains in the simulation log.

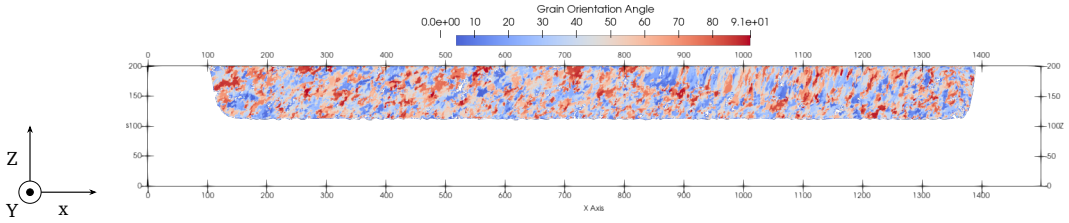


Figure 4.1: Single melt track with nucleation $N_0 = 0$, where all growth stems from existing grains along the melt pool boundary.

To test the validity of the temperature-based state transition, simple scenarios with a $T_{\text{liq}} > T_{\text{max}}$ and $T_{\text{sol}} < T_{\text{min}}$ were tested, resulting in no liquid cells and no solid cells, respectively.

A performance-critical simulation parameter is the minimum distance below the current lasing height that will still be calculated. As detailed in Section 3.1.4, this method relies on extracting a domain $\mathcal{D}$ to simulate at every time-step, with the assumption that no cell transitions would occur outside this domain. This has been verified visually with simulations using larger active domains $\tilde{\mathcal{D}}$, inspected in a slice view in ParaView, where no changes were detectable at the edges of $\mathcal{D}$. The simulation times increased proportionally to an increase in $\mathcal{D}$, meaning a $\tilde{\mathcal{D}}$ twice the height of $\mathcal{D}$ resulted in a roughly doubled simulation time. This was also verified by selecting a $\bar{\mathcal{D}}$ half the height of $\mathcal{D}$, roughly equal to the maximum depth of the melt pool, where the average grain size at each height level varied considerably in comparison to the larger domains, as seen in Fig. 4.2.

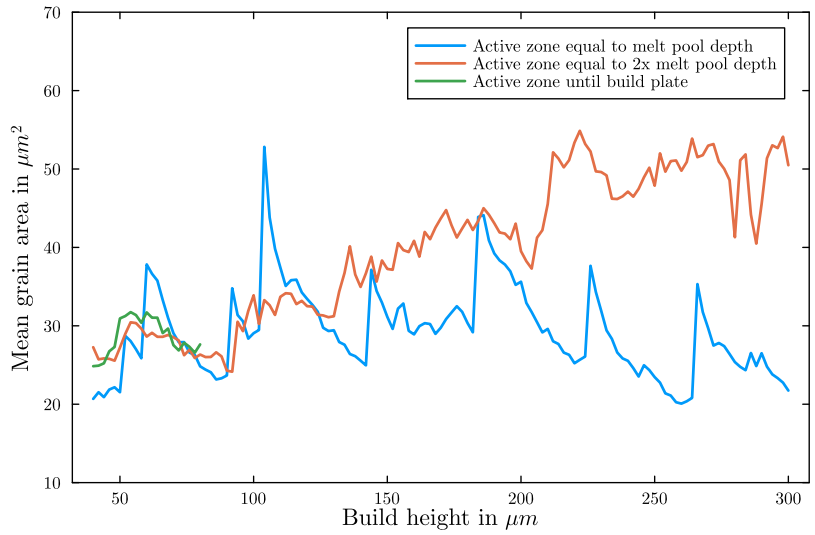


Figure 4.2: Comparison of different sized active zones, with the mean grain area calculated for every slice in X-Y plane. The full simulation without any extraction of active areas was too computationally costly to simulate for the entire height but is included for completeness.

4.1.3 Convergence Analysis

A convergence analysis was performed for the single-track scenario to verify the chosen cell size Δx for further calculations. This was performed as described in Section 3.1.5, obtaining results as seen in Fig. 4.3. The high values for $\Delta x = 10\ \mu m$ stem from the lack of grains larger than 1 cell, where an ellipsoidal approximation was impossible to calculate. The values converge for 2 μm, and whilst it would have been ideal to calculate this for 1 μm, this was not possible due to memory constraints, as a mesh this size would have required well over 100GB of RAM due to the inverse cubic proportionality between array sizes and mesh distances.

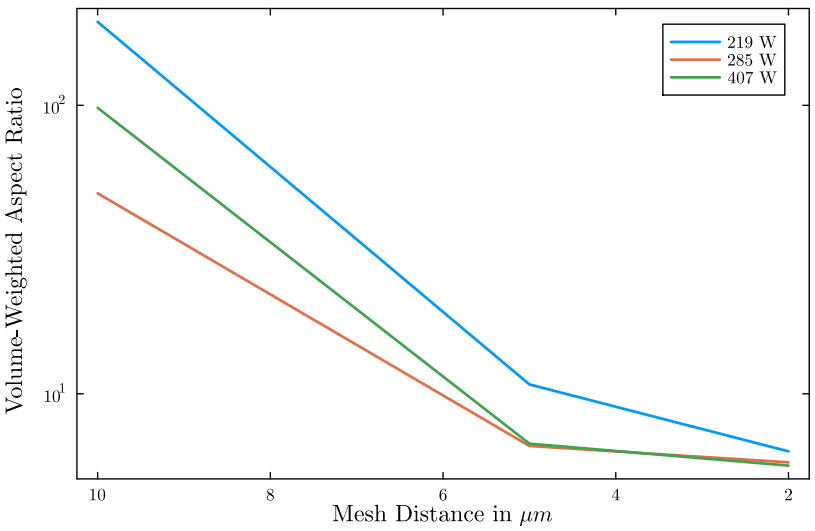


Figure 4.3: Convergence Analysis for mesh distances Δx in the CA mesh.

4.1.4 Performance

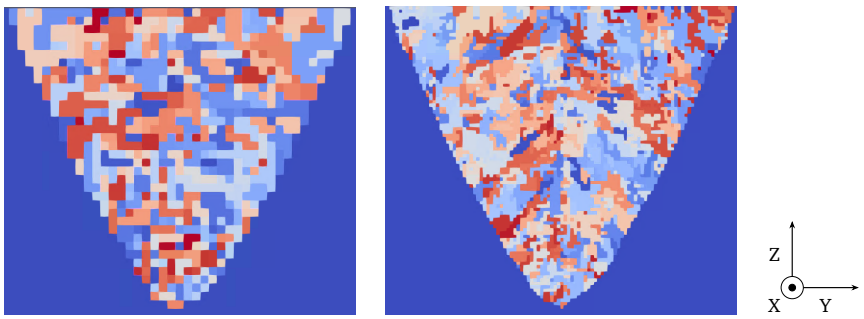
Whilst not the focus of the thesis, a high computational performance is advantageous for practical use. For the simulated domain with $n_x = n_y = 300$ and $n_z = 250$, with physical dimensions 0.6 x 0.6 x 0.5 mm, the computation time was typically around 12h. As few calculations are performed in the simulation, the problem is clearly data-bound. Increased computation times for higher mesh resolutions mainly stem from slow interpolation of the temperature field, and when extracting the indices for which calculations are performed. This was analyzed using the log output, as computation times for all steps are logged. Further efficiency improvements for very large mesh sizes should be focused on reducing the amount of searching across the arrays, and rather pass indices between function steps.

4.2 Validation of Microstructure Simulation

Once it is verified that all components of the simulation work as intended, validation is critical. The validation procedures for the microstructure outlined in Section 3.1.5. First, qualitative measures are utilized to validate the microstructure evolution against known physical phenomena before quantitative comparisons are performed to validate the multi-layer growth and how well the simulation generalizes across different temperature inputs without individually adjusting parameters for every scenario. The parameters used all numerical and qualitative results are obtained with the material parameters described in Table 3.3 and simulation parameters from Table 3.1 unless otherwise specified.

4.2.1 Qualitative Comparison

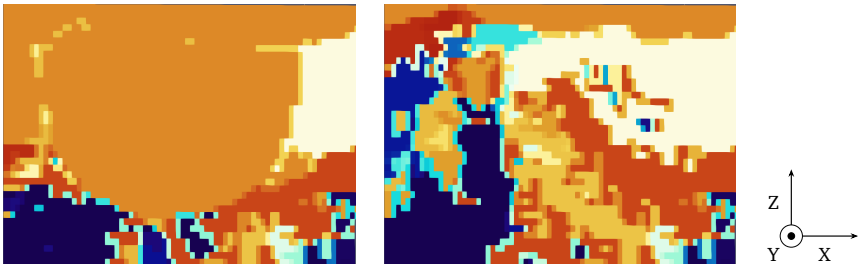
For the qualitative comparisons, some slices of the simulation output are visualized with ParaView to illustrate how well different grain growth phenomena are modeled. The solidification behavior of melt pools was investigated for both a single laser melt track and for arbitrary melt pools in the multi-layer build. For the single melt track, this is illustrated in Fig. 4.4 for varying mesh distances, corresponding to the scenarios calculated during the convergence analysis.



(a) Δx = 5 μm **(b)** $\Delta x = 2\ \mu m$

Figure 4.4: Solidification of a melt pool sliced normal to scan direction, with a total melt pool width of 220 μm at the surface on top.

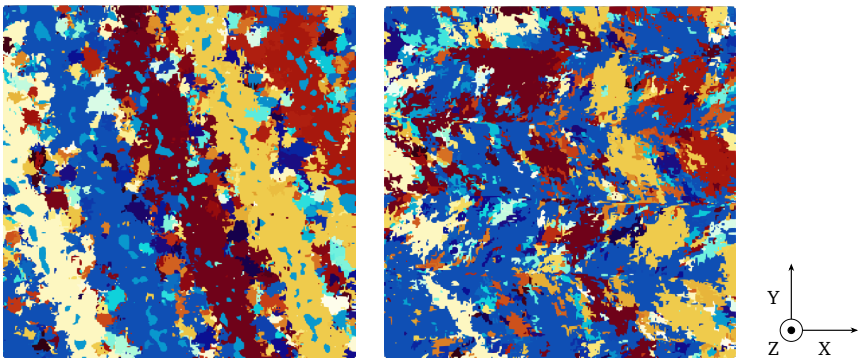
For the multi-layer simulation, the growth of neighboring grains into a melt pool is seen in Fig. 4.5. As the melt pools are much narrower in the multi-layer simulation, the grain growth is considerably coarser than in the single-layer simulation when using the same mesh with width $\Delta x = 2\ \mu m$.



(a) Before Solidification **(b)** After Solidification

Figure 4.5: Solidification of a melt pool sliced normal to scan direction, with a melt pool width of 100 μ m.

The effect of remelting when the laser scans across higher layers, as seen in Fig. 4.6 can also be used for validation. Whilst we can verify that the remelting takes place, the finding in MOSTAFA et al. (2017) was that the grain size was larger in regions molten twice than in those molten only once. This could not be confirmed in general. A range of different nucleation parameters were tested, and with a nucleation rate $N_0 = 6e7$, the effect could be observed, but with $N_0 < 1e7$, this effect was no longer visible, however a higher nucleation rate was not found viable by qualitative comparison with experimental EBSD images demonstrating grain sizes best reflected by $N_0 = 3.5e6$, as used in the simulations.



(a) First solidification **(b)** Solidification after remelting

Figure 4.6: Effect on grain structure in a X-Y layer, perpendicular to the BD, being remelted when the laser scans across the next layer. For Simulation Scenario 2 with 285W laser power.

Another important validation aspect is the columnar grain growth across layers in the PBF-LB/M build. An example of this is seen in Fig. 4.7, where we can see that grain grows across several layers. This verifies that it is possible with the current implementation for grains to grow across layers, but does not correlate well with experimental results that typically show a much clearer growth preference in the BD, and rarely see the growth in two perpendicular directions, marked by the smallest circle in the figure. Also note the clear difference in grain sizes across different areas close to each other within the same build layer. The locations marked by the largest two circles exhibit a very different structure, where the smaller one has several grains of similar size, all oriented roughly in the BD, whilst the larger circle shows an area where three very large grains dominate the structure, with grain areas well

above 1000 *mu*m. Again, this stems from the nucleation modeling, permitting the nucleation of very few grains in some regions, typically in the middle of the part seen in the studied scenarios, and the nucleation of many new grains in other regions.

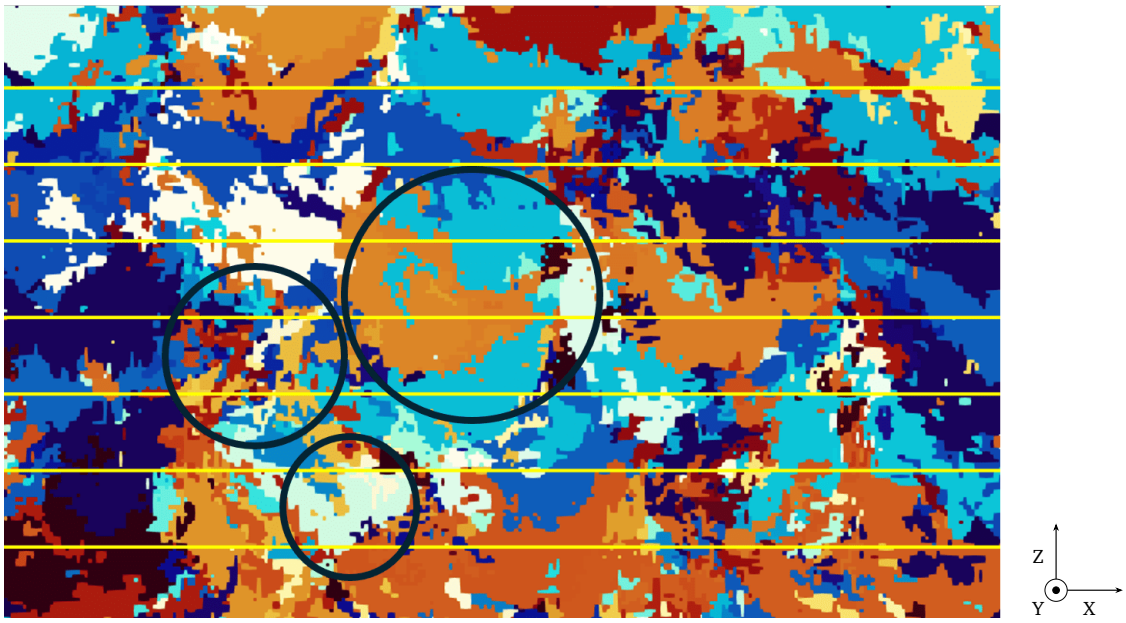


Figure 4.7: Grain structure in the X-Z plane cut in the middle, with the powder bed height at each layer marked in yellow. Output of Simulation Scenario 2 with 285 W laser power.

4.2.2 Quantitative Comparison

It has been demonstrated that several microstructural phenomena are not observable in the simulation output of individual simulation outputs, and the same can be expected for the quantitative comparison across several simulations. For a quantitative validation, experimental results by ZHAO (2024) are compared to the simulation outputs of corresponding simulations. Due to time constraints on the simulations, only the bottom volume, dubbed "Scenario 0," and the top volume, dubbed "Scenario 2," have been investigated, illustrated in Fig. 3.7. To obtain grain areas appropriate for comparison, the volume has been sliced into 100 evenly spaced slices, and the area has been determined by a count of cells belonging to each grain in the slice. From this, heterogeneity within the part can be quantified and gives the value used for comparison with the experimental results when averaged over all slices. The outer 10% of each side length has been excluded from these calculations to obtain values representative of the inner part of the geometry and avoid the effects of bad boundary conditions in the temperature simulation.

In Fig. 4.8 a comparison between the simulated and experimental results is shown, where one can clearly see that the simulation does not manage to capture the higher grain sizes for increasing laser power. However, Scenario 2 exhibits considerably higher temperatures than Scenario 1, and the increased grain size due to this is reflected in the simulation output. It is hypothesized that an increase in grain size due to increased heat flow is modeled correctly, but hindered across layers due to the excessive growth of freshly nucleated grains in the current layer, limiting the effect of increased heat flow.

When considering the average grain sizes in the X-Y plane, the correlation between grain size and laser power is non-existent and shows a trend opposite to the experimental results.

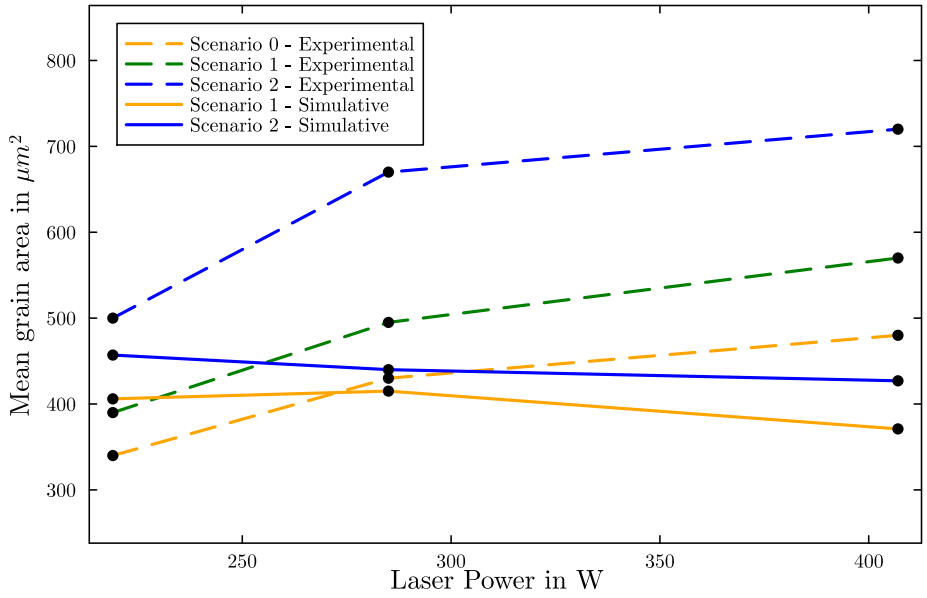


Figure 4.8: Comparison of simulated data in the Y-Z plane and experimental data from ZHAO (2024).

This same pattern holds for the total grain density across the whole built part, indicating that the nucleation model in its current implementation exhibits highly heterogeneous nucleation densities and consistently models higher nucleation rates for steeper temperature gradients. The comparison with the experimental results should not be assigned too much weight, as the experimental investigation considered only the grain areas parallel to the BD.

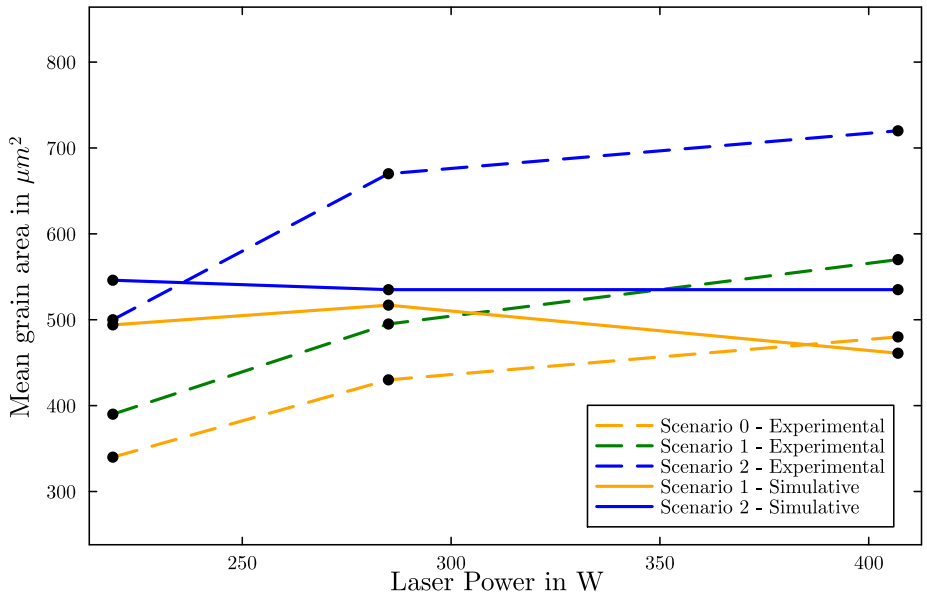


Figure 4.9: Simulated data in the X-Y plane and experimental data in the Y-Z plane from ZHAO (2024).

4.3 Fracture Model

The microstructure-aware fracture model described in Section 3.3 is calibrated in this section, limited to experimental results from available papers as fracture experiments are out of the scope of this thesis.

The extended fracture model gives a critical strain ε_e^f, where the material fails and cracking initiates. This critical strain at failure is defined as:

$$\varepsilon_e^f = (d_1 + d_2 \exp(d_3 \sigma^*)) \left(1 + d_4 \ln \dot{\varepsilon}^*\right) (1 + d_5 T^*) (1 + d_6 \delta^*)(1 + d_7\, \delta^* sin^2(2(\vartheta^* + d_8))) \,, \quad (4.1)$$

where the parameters d_1 to d_5 are adopted from PANZER (2020) which determined that the JC model was suitable for IN718 and determined these parameters experimentally. Parameter d_6 should be determined from fracture experiments with loading normal to the BD, which was investigated in PANZER et al. (2024). Parameters d_7 and d_8 should be determined with fracture experiments at 45° angles towards the BD, for which no experimental results were found for IN718. BRENNE and NIENDORF (2019) performed comprehensive experiments on 316L stainless steel, the validity of which cannot be assumed for IN718. Due to a lack of available experimental results, the parameters d_7 and d_8 are neglected, meaning $d_7 = d_8 = 0$.

During the fracture experiments in PANZER et al. (2024), the crack length L_c and grain size A_g where measured for differing laser powers. By performing a linear regression analysis on the experimental results, a linear correlation between the parameters could be obtained, as seen in Fig. 4.10.

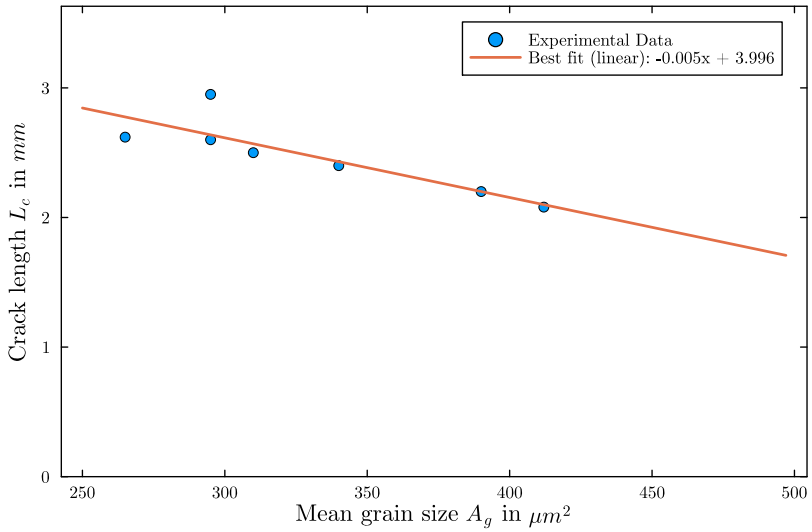


Figure 4.10: Results of a linear regression analysis on experimental data from PANZER et al. (2024).

Rewriting the linear regression in terms of δ^* we obtain:

$$L_c(A_g) = L_{\mathrm{c,ref}} - 1.26\delta^* \,, \quad (4.2)$$

where $L_{\mathrm{c,ref}}$ is the crack length for the smallest grain size, and δ^* is the relative grain size, as defined in Section 3.3. However, to calibrate d_6 a correlation between the strain at failure ε_e^f and the grain size A_g is needed. A relationship between these parameters can be inferred using Eq. 2.5, assuming elastic deformation:

$$\varepsilon_e^f \propto -\sqrt{L_c} \,. \quad (4.3)$$

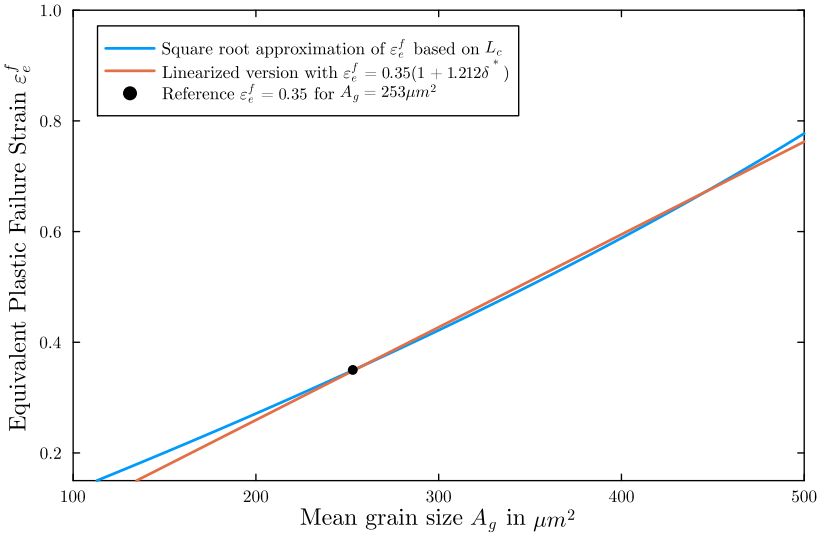


Figure 4.11: Results for a linearized model of fracture model factor accounting for grain sizes based on the experimental data from PANZER et al. (2024) and PANZER (2020)

As this only gives proportionality and has some limiting assumptions regarding linear-elastic fracture behavior, further experiments should be conducted to better quantify this relationship. Calibration will still be attempted, assuming that the smallest grain sizes have a ε_e^f corresponding to Specimens 2 and 3 from PANZER (2020), Section 5.5, used for fitting the JC model. With an initial $\varepsilon_e^f = 0.35$ and assuming the proportionality constant for Eq. 4.3 is 1, ε_e^f has the following dependence on the grain size:

$$\varepsilon_e^f = C - \sqrt{L_{\mathrm{c,ref}} - d_{6,\mathrm{nl}}\delta^*}\,, \tag{4.4}$$

where $d_{6,\mathrm{nl}} = -1.26$ and $C = 2$ is some constant. As seen in Fig. 4.11, the error using a linear approximation is negligible, with a linear approximation,

$$\varepsilon_e^f = C(1 + d_6\delta^*)\,, \tag{4.5}$$

where $d_6 = 1.212$ and $C = 0.35$ represents the JC model calibrated for the scenario.

4.4 Limitations of Presented Models

There are several limitations of this simulative approach that need to be addressed. As mentioned in Section 2.4, the combination of physical and heuristic approaches in the crystal simulations necessitate both material parameters and manual calibration. This section will address what limitations this poses for the validity of the results and what efforts will be required to apply this method and code to similar problems.

4.4.1 Limitations of Microstructure Simulation

What seems to be the greatest weakness of the model in its current form is the implementation of the nucleation model. Whilst the simulation captures the dependence of the grain

density on heat flow and the variation in grain density between regions experiencing single or double-pass melting, the microstructure on a qualitative level does not exhibit the expected structure. Although columnar growth is visible, the growth of columnar grains across several layers, characteristic for PBF, seems to be impeded by the nucleation of new grains in subsequent layers. The nucleation model employed also cannot accurately predict the grain density for differing laser powers, as a lower laser power seems to permit less nucleation, contrary to experimental results. KOEPF et al. (2018) devised an alternate approach where nucleation is not modeled; instead, pre-initialized grains in the structure are activated during solidification. This seems to solve the problem of irregular grain density observed in this model, but knowledge about the expected grain density for given process parameters is required. LIAN et al. (2019) determined realistic microstructures using a combination of bulk nucleation and epitaxial growth by reactivation of adjacent cells, both of which are implemented in this thesis as well, but it was not possible to find any satisfactory combination of simulation parameters to achieve similar results.

The nucleation process with the bulk nucleation model requires significant tuning before it produces the desired results for a given process parameter. As mentioned by RAPPAZ and GANDIN (1993) in the original paper on the model, it is important that the nucleation rate remains low, and always below unity. The only way to enforce such a constrain in this project, was found by relating the adaptive time-stepping to the nucleation rate. As the nucleation rate is modeled by the integral over a normal distribution, any undercooling approaching ΔT_c, the mean undercooling at nucleation, will produce excessive nucleation. The solution in this thesis has been to decide on a maximum allowed nucleation rate per step of crystal simulation and set the timesteps such that $N(max(\Delta T_{qc}))$ is below that value.

$$\Delta T_{qc} = T_{qc}^{t-1} - T_{qc}^{t} \,, \tag{4.6}$$

where T_{qc} is the temperature of cells that have cooled from above T_{liq} to below T_{liq} in the current time-step.

Some variant of temperature-dependent time-stepping is also necessary for a functioning algorithm, as there is no process available to solidify a cell properly if $\Delta T_c > \Delta T_{max} = T_{liq} - T_{sol}$, as the cell never becomes mushy, and has no opportunity to nucleate or become captured by an adjacent grain. This functionality may be to blame considering the issues with the nucleation model, so changing the heuristic for the adaptive time-stepping should be a priority beyond this thesis. The high spatial heterogeneity of nucleated grains is observed in Fig. 4.12, where the small circle indicates a region where a melt pool was solidified almost exclusively by nucleation, rather than grain growth from surrounding grains. The large circle highlights an area typical of too high nucleation rates, where the growth seems to be modeled well within the layer, but growth across layers cannot occur due to excessive nucleation of new grains blocking epitaxial growth.

Beyond the lack of physical correctness of the simulation output, the implementation also has limitations on the simulation domain. As no adaptions for distributed computing have been made, the size of simulation domains is limited by RAM available on a single computation node. This should be easy to implement using the DistributedArrays.jl library, with little overhead, as the algorithm depends on only a very small neighborhood around the cell to perform a simulation step. The current implementation can handle arbitrary geometries but does not consider this explicitly in the mesh, meaning the simulation domain will depend on the bounding volume size of the simulated geometry.

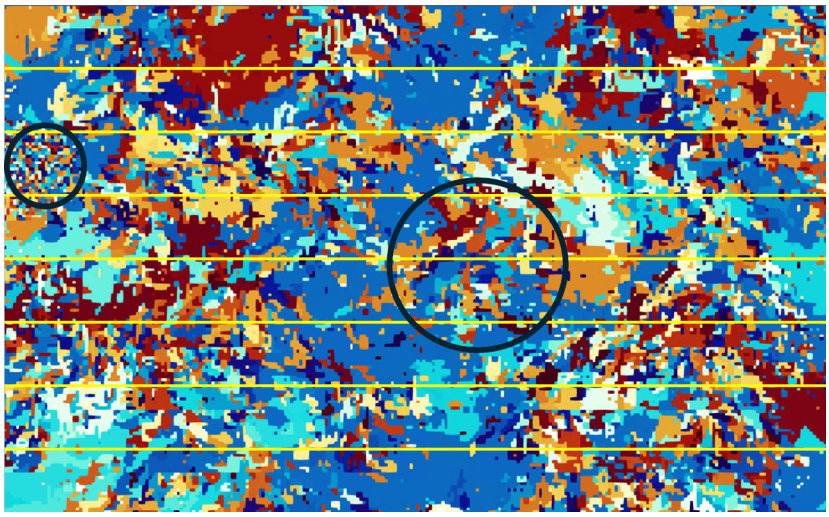


Figure 4.12: Grain structure in the X-Z plane cut in the middle, with the powder bed height at each layer marked in yellow. Output of Simulation Scenario 0 with 407 W laser power and $N_0 = 6e7$.

4.4.2 Limitations of Fracture Model

The employed fracture model is purely empirical, which results in some clear limitations. The validity can only be assumed for the materials, microstructure sizes, and loading directions for which it has been calibrated. When implemented numerically as part of fracture modeling that employs the model as a criterion for fracture initiation, it relies on manually calibrated microstructure abstraction methods. As it is currently implemented, it is necessary to manually determine cutoffs for what is considered a statistical outlier using histograms, and the regions for which the grain size and angles should be calculated will also vary depending on the size of the crack tip and plastic zone around the crack tip. The results obtained in Section 4.3 should be considered carefully and will have to be verified by further experiments to ensure a correlation between the crack length L_c and the strain at failure ε_e^f, or calibrated directly with new experiments.

Chapter 5

Summary and Outlook

5.1 Summary

This project aimed to implement a functional microstructure simulation and use this information to determine estimates of the fracture toughness in parts built using PBF-LB/M. For this purpose, a simulation code was developed, integrated with an existing FEM simulation to simulate the layer-wise melting of metal powder and its subsequent solidification by grain nucleation or grain growth. This necessitated a 3D simulation, where the Decentered Octahedron CA-based algorithm used widely in literature was chosen. This numerical model has been shown capable to simulate competitive grain growth, which is an important phenomenon in PBF-LB/M, without incurring prohibitive computational costs. Verification procedures were followed, showing a correct implementation of the algorithm, and validation was performed by comparing the results qualitatively and quantitatively with experimental results. The validation showed that the columnar grain growth known to occur across build layers in PBF-LB/M was often halted by excessive nucleation in a new layer, or grew in several directions due to lack of competing grains. Other phenomena, such as remelting of lower layers, grains angled towards the scan direction, clear grain boundaries between adjacent melt tracks, and larger grain sizes for higher heat flows, were successfully modeled. The simulation model also could not accurately predict the grain density when process parameters were changed in the simulation, where increased laser power led to smaller grains rather than larger grains, possibly due to the strong dependence of the nucleation model on the temporal thermal gradient. The simulation code developed can work stand alone or coupled with ongoing FEM simulations and is capable of simulating the grain growth in arbitrary geometries. The computation times for the chosen mesh size $\Delta x = 2\ \mu m$, and a simulation domain of 0.6 x 0.6 x 0.5 mm, were below 12h. The thermal FEM simulations, however, required 48h for the same domain, meaning coupled CA-FE simulation can be performed with limited additional cost.

To estimate the fracture toughness an extension of the JC fracture model was devised, based on experimental results from previous papers. This extended the fracture model to account for grain sizes and anisotropic grain structures, where fracture toughness increases with higher mean grain sizes, and for cracks at a 45° towards large columnar grains. The dependence on mean grain sizes was calibrated using existing experimental results, whilst the modeling of the anisotropy still requires further experiments to calibrate the parameters. No experimental validation of the fracture model was performed within the scope of this thesis.

5.2 Outlook

As noted at several points during the thesis, there is considerable need for further work to improve and validate the presented methods, a summary of which is presented here.

5.2.1 Improving the Microstructure Simulation

The microstructure simulation model in its current form can successfully simulate the microstructure of single melt tracks from coupled FEM simulation, but struggles with capturing the process parameter dependence of grain nucleation and growth of columnar grains in multi-layer simulations. Further work to fine-tune or rework the existing nucleation model is deemed necessary for results that can be employed for fracture estimates. LIAN et al. (2019) also simulated multi-layer AM builds using the here employed bulk-nucleation, but used additional adjustments to ensure columnar growth.

5.2.2 Experimental Calibration and Validation of Fracture Model

The proposed microstructure-aware fracture model requires data from several specialized experiments to quantify the anisotropy and grain size dependence of fracture toughness. Experiments for calibration of this model, and subsequent validation of the model performance should also be performed.

5.2.3 Alternative Fracture Models

The empirical fracture model proposed in Section 3.3 only uses statistics of the simulated microstructure, and relies on calibration experiments to quantify the model parameters. A physics-based fracture model that utilized the entire simulated microstructure information would be preferable, as it could quantify heterogeneities in the fracture toughness and rely on a rigorous mathematical derivation from fracture mechanics. CPFEM is a state-of-the-art numerical method integrating grain-level microstructure information into FEM calculations and has been demonstrated to work for fracture problems on conventionally manufactured metal components. It has been used for fracture calculations in conventionally produced metals (SIDDHARTH et al. 2024), and for assessment of mechanical properties in AM (CHAKRABARTY et al. 2023).

Appendix A

Appendix A

A.1 Functions

A.1.1 Pre-processing:

calcBarycentricCoords: Calculates the factors for barycentric interpolation between the coarse tetrahedral mesh, and the significantly finer cartesian CA mesh. A bounding box is determined for every element, and the factors determined by Cramer's rule. The factors, and which node they belong to are stored to baryCoordsArr and elemArr in HDF5 format. Based on ERICSON (2005), extended to 3D. Resolutions should be chosen such that the smallest tetrahedron contains 4 points to ensure that all node temperatures are accounted for.

preInitializeMesh: Initializes the crystal mesh in a Voronoi-like fashion, where a number of grains are randomly scattered within the volume, and all cells within some random close neighborhood are assigned to that grain. The remaining cells are assigned using the *updateInterfaceCell* function.

getConfig: A script that reads a .toml config file and the input geometry, determining the remaining simulation parameters which are again stored as a .toml file, and supplied to the simulation.

A.1.2 Per-step-processing:

getActiveArea: Determines the region of interest for the simulation according to Eq. 3.16, and supplied the crystal growth functions with a reduced computational domain $\widetilde{\mathcal{D}}$.

getStepSize: A function determining the necessary substeps between supplied temperature input to satisfy a CFL-like condition for the dendrite growth, and a chosen maximum undercooling for a cell that was liquid in the previous step.

tempBaryInterpolation: Barycentric interpolation from node temperatures from FEM simulation to a cartesian mesh, based on interpolation factors from calcBarycentricCoords.

tempLinInterpolation: Interpolation with linear splines using Interpolations.jl, up from the minimum viable resolution, to the desired mesh resolution for the CA calculations. No loss of accuracy compared to a higher resolution in calcBarycentricCoords, provided linear tetrahedral elements are used in the FEM.

tempTimeInterpolation: Linearily interpolates the temperature field within $\widetilde{\mathcal{D}}$ between the previous and the current time step.

A.1.3 Crystal Growth:

updateCellStates: Updates all cells according to the folowing update transition rules in Eq. 3.2.

nucleateCells: A phenomenological nucleation integral 2.22 is calculated with QuadGK using adaptive Gauss-Kronrod quadrature) for all mushy cells. If it is larger than a random number in (0,1), a grain nucelates at this cell.

updateCellInterfaces: Updates all mushy cells directly adjacent to a solid cell to be a part of the solid grain.

UpdateDendriteLenghts: Updates the dendrite lengths in cellOctaArray for all active solid cells according to Eq. 2.21, as a model for dendrite growth.

UpdateCellGrowth: Finds all mushy cells, and their solid active neightbors in a 2nbd moore neighborhood. Iterates in randomized direction within each neighborhood in 1n, 1m, 2n, 2m order, stopping at the first solid cell that fullfills the capture requirements according to the decentered octahedron algorithm described in 3.1.2

engulfCell: Calculated the grain centre and the lengths to initialize a new octahedron.

Interfacing with C

julia_interface.c: Loading of Julia functions that need to be accessible from C.

functionsToC.jl: Exposes some Julia functions to C, and defined I/O data types.

loadMesh: Retrieves appropriate interpolation factors, simulation parameters and dimensions for further use.

crystalGrowthC: Handles appropriate loading and storing of arrays before performing crystal growth calculation.

getArray: Loads a limited domain $\widetilde{\mathcal{D}}$ of a 1D array from C to Julia.

storeTemps: Stores temperatures in HDF5 format, for later use in standalone simlations.

Exporting Data

exportLogVars: Handles logging to a .CSV file.

exportVisualization: Stores the supplied 3D scalar or vector data in appropriate rectilinear grid .vtr files, with a .pvd metadata file for easy inspection and visualization in Paraview using the WriteVTK module.

exportH5: Handles exporting of data to HDF5 format using the HDF5.jl module.

Data Extraction

getEllipse: Calculates elliptic envelope of grain using an approach outlined in Section 3.2, based on McElfresh et al. (2024). Discards the 10% of points belonging to a grain with the

furthest distance from the ellipse centre to be more resistant to outliers. Returns both the axis lengths, directions, and raw counts of number of cells belonging to a grain.

getEllipsoid: 3D analog of getEllipse.

getSlicedGrainSizes: Iterates over all slices along a chosen axis to determine the distribution of average grain sizes over either build height, or distance to the border of the build.

A.2 List of Programs

The following programs were used in this work:

- CalculiX CrunchiX 2.20 (DHONDT 2004)
- Julia 1.10.3 Julia Programming Language (BEZANSON et al. 2017)
- ParaView 5.12.0 (AHRENS et al. 2005)

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